STUDY ON LOW-GWP ALTERNATIVE TECHNOLOGIES FOR CHILLERS IN LARGE AIR CONDITIONING BUILDINGS

OZONE CELL

MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE
GOVERNMENT OF INDIA

STUDY ON LOW-GWP ALTERNATIVE TECHNOLOGIES FOR CHILLERS IN LARGE AIR CONDITIONING BUILDINGS

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OZONE CELL

Ministry of Environment, Forest and Climate Change (MoEF&CC)

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भारत सरकार

सत्यमेव जयते

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BHUPENDER YADAV

MINISTER
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MESSAGE

India's growing air-conditioning sector is driven by critical factors such as energy efficiency requirements, climate friendly technologies, market expansion, and green initiatives. These advancements in the cooling technologies of commercial and industrial buildings are crucial towards sustainable climate solutions.

Chillers are a preferred choice for cooling of large buildings because they provide a centralized and efficient way to meet significant cooling needs. Well-designed chiller systems can significantly reduce cooling energy consumption and electricity costs, besides promoting energy efficiency and productivity by managing heat loads.

Low Global Warming Potential (GWP) technologies in chiller sector provide positive environmental impact, meeting compliance to the Montreal Protocol and its Kigali Amendment and domestic regulations, leading to lower operational costs. They also promote sustainable practices within the cooling industry.

The study on low GWP alternative technologies for chillers in large air conditioning buildings assesses the feasibility of adoption of low GWP technologies to meet the cooling needs of large buildings. I congratulate all those involved in the preparation and consultation of this report.

(Bhupender Yadav)

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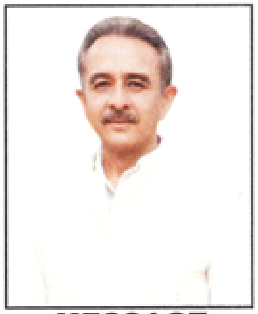

MESSAGE

Chillers cater to the cooling needs of large-scale buildings like commercial buildings, hospitals, and data centres that generate significant heat due to their capacity to handle high cooling loads throughout the building with precise temperature control.

Current chillers using high global warming potential (GWP) hydrofluorocarbon (HFC) refrigerants need to be phased out in view of the Montreal Protocol commitments. The Kigali Amendment to the Montreal Protocol lays emphasis on adoption of low global warming potential alternative technologies during the phase down of HFCs and accordingly, the trend is shifting towards low-Global Warming Potential (GWP) alternatives, including for chiller applications leading to significant environmental benefits.

Innovation in design of chiller compatible to low GWP refrigerants, training for installers and maintenance personnel to handle such refrigerants and systems safely, promoting awareness on adoption of low GWP technologies will lead towards environmentally friendly cooling for large buildings

The publication on low GWP alternative technologies for chillers in large air conditioning buildings would serve as an important resource material and should be disseminated widely amongst all concerned stakeholders.

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(Kirti Vardhan Singh)

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AND CLIMATE CHANGE

MESSAGE

Growth of cooling demand in buildings has been rising steadily over the last decade due to a combination of factors, such as rising population living largely in tropical urban clusters, growing aspirational needs fuelled by sustained economic growth. As per the India Cooling Action Plan (ICAP), it is estimated that 70% of the new building stock is to be developed in the coming years. It is therefore essential to incorporate energy efficient and cost-effective cooling systems through which buildings can have inherently reduced energy consumption over its operating lifetime.

Chiller based cooling systems enhance thermal comfort by removing heat from large buildings, producing cool air to maintain optimal temperature and humidity levels for its occupants. Efficient chiller systems, combined with smart control strategies and passive building design, supplemented with low Global Warming Potential (GWP) and energy efficient refrigerant technologies promote sustainable cooling and balance occupant satisfaction.

The Study on low-GWP alternative technologies for chillers in large air conditioning buildings will be useful in promoting energy efficient and cost-effective cooling solutions. I compliment the team associated with the preparation of this report.

(Tanmay Kumar)

Place: New Delhi
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Abbreviations

AEEE	Alliance for an Energy Efficient Economy
ARI	Air-Conditioning and Refrigeration Institute
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BEE	Bureau of Energy Efficiency
BEMS	Building Energy Management Systems
BIS	Bureau of Indian Standards
BMS	Building Management Systems
CHP	Combined Heat and Power
COP	Coefficient of Performance
DCS	District Cooling System
DEC	Direct Evaporative Cooling
DST	Department of Science & Technology
ECBC	Energy Conservation Building Code
EER	Energy Efficiency Ratio
GPP	Green Public Procurement
GRIHA	Green Rating for Integrated Habitat Assessment
GWP	Global Warming Potential
HCFC	Hydrochlorofluorocarbon
HFC	Hydrofluorocarbon
HFO	Hydrofluoroolefin
HPMP	HCFC Phase-Out Management Plan
ICAP	India Cooling Action Plan
IEC	Indirect Evaporative Cooling
IoT	Internet of Things
IPLV	Integrated Part Load Value
LEED	Leadership in Energy and Environmental Design
LiBr	Lithium Bromide
LNG	Liquefied Natural Gas
MoEF&CC	Ministry of Environment, Forest and Climate Change
NBC	National Building Code
ODP	Ozone Depletion Potential
OEM	Original Equipment Manufacturer
PAT	Perform, Achieve and Trade
PCM	Phase Change Material
PDRC	Passive Daytime Radiative Cooling
PV	Photovoltaic
SDA	State Designated Agency
SEZ	Special Economic Zone
TERI	The Energy and Resources Institute
TES	Thermal Energy Storage
TR	Tonnage of Refrigeration
TRL	Technology Readiness Level
ULB	Urban Local Body
UNEP	United Nations Environment Programme
VCR	Vapor Compression Refrigeration
VRF	Variable Refrigerant Flow

INTRODUCTION

The demand for space cooling in India has witnessed a significant increase over the past decade, driven by rapid urbanization, rising ambient temperatures, and the expansion of commercial, institutional, and public infrastructure, leading to increased demand for cooling. Addressing the rising cooling requirement is both a challenge as well as a unique opportunity. As per the India Cooling Action Plan (ICAP), the aggregated nationwide cooling demand, in Tonnage of Refrigeration (TR), is projected to grow around 8 times by 2037-38 as compared to the 2017-18 baseline [1–3]. The building sector cooling demand shows the most significant growth at nearly 11 times as compared to the baseline. Another important aspect to note is that a large part of the cooling demand is catered through refrigerant-based cooling. Refrigerants used in cooling equipment are regulated under the Montreal Protocol. For cooling requirements of large-scale buildings, chillers are feasible and widely used because they are highly energy-efficient, provide consistent cooling, and can handle large cooling loads more effectively than conventional air conditioning systems. Chillers, along with their supporting auxiliaries comprising pumps, cooling towers, and air-handling units, represent the most energy-intensive component of these systems. Their design, operation, and refrigerant choice therefore play a decisive role in shaping both energy demand and climate impact [4–7].

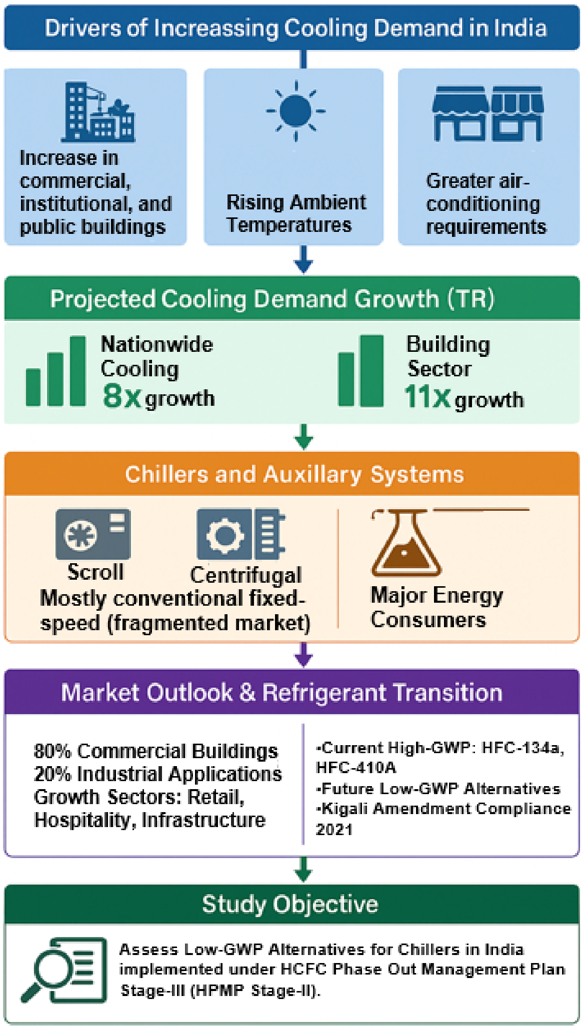


Fig. 1.1 Cooling demand and Low-GWP transition in India.

India’s chiller market is characterized by a wide range of technologies scroll, screw, and centrifugal chillers catering to different capacity segments. While screw and centrifugal chillers have incorporated advanced features such as high-efficiency heat exchangers, flooded evaporators, and oil-free variable-speed compressors, the scroll chiller segment remains fragmented, with many small-scale manufacturers still deploying conventional fixed-speed technologies. The centralized air conditioning building sector accounts for nearly 80% of chiller sales in India, with the remaining 20% deployed in industrial applications. Market growth is projected to remain strong over the next decade, particularly in the retail, hospitality, and infrastructure sectors, with screw and centrifugal chillers expected to dominate future installations due to their higher efficiency. At the same time, India is undergoing refrigerant transition to low global warming potential options in line with the compliance obligations under the Montreal Protocol. Having ratified the Kigali Amendment in 2021, India will have to plan and strategize phase down of HFC-134a and HFC-410A, currently used high GWP refrigerants in the Chillers and transition to viable low GWP alternatives, considering the challenges of availability and accessibility, safety and competencies of the personnel associated with maintaining and servicing the equipment [8–10].

The Study on Low-GWP Alternative Technologies for Chillers in Large Air-Conditioned Buildings” aims to assess the low GWP options for chillers in the Indian context (Fig. 1.1). This study is being undertaken as part of the enabling activities under the HCFC Phase-Out Management Plan Stage-III (HPMP Stage-III), implemented by the Ozone Cell of the Ministry of Environment, Forest and Climate Change (MoEF&CC), Government of India, in close cooperation with UNEP as the implementing agency.

2. OVERVIEW OF CHILLERS IN AIR CONDITIONING APPLICATIONS

In India's rapidly growing urban landscape, large commercial buildings such as hospitals, shopping malls, airports, and IT parks rely heavily on central air-conditioning systems, with chillers forming the core of their cooling infrastructure (Fig. 2.1).

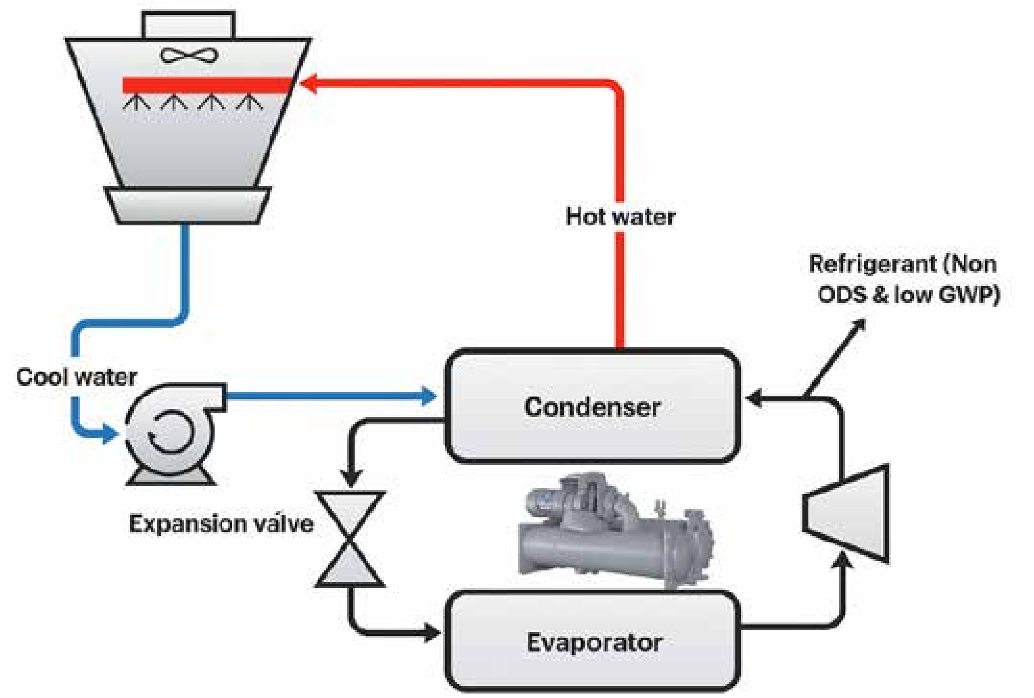


Fig. 2.1, Vapor compression chiller system.

These systems account for a considerable share of electricity use in the built environment and contribute significantly to direct and indirect greenhouse gas emissions. Traditionally, chillers in India are operated using refrigerants such as HCFC-22 and HFC-134a, which have high Global Warming Potential (GWP). With the implementation of the Montreal Protocol and its Kigali Amendment, there is now a clear mandate to phase down such substances. India has already phased out use of HCFC from manufacturing of new equipment including chillers as on 1.1.2025. The transition to environmentally friendly alternatives with low GWP need to be explored. Refrigerants such as R-1234yf, R-290 (propane), R-717 (ammonia), CO_2, and R-32 are emerging Low GWP alternatives. Each brings its own set of advantages in terms of efficiency and environmental impact, but also presents challenges relating to flammability, toxicity, system redesign, and cost. [11]

This study, therefore, assess currently available low GWP technology options for chiller applications. Presently the following low GWP refrigerants options are available for Chillers applications.

i. Types of Low-GWP Refrigerants

The gradual phase-out of ozone-depleting and high-GWP refrigerants has shifted global focus toward alternatives that are both environmentally responsible and technically viable. In the context of centralized air-conditioning systems, especially in large buildings, three broad categories of low-GWP refrigerants have emerged as leading options:

hydrofluoroolefins (HFOs), natural refrigerants, and next-generation hydrofluorocarbons (HFCs). Each category presents its own set of trade-offs in terms of performance, safety, compatibility, and market readiness (Table 2.1).

a) Hydrofluoroolefins (HFOs)

HFOs are a new class of synthetic refrigerants designed with significantly lower climate impact than earlier HFCs. Their thermodynamic behavior is similar to conventional refrigerants, allowing easier integration into modern chiller systems.

- R-1234yf and R-1234ze are among the most established HFOs. Both have a global warming potential (GWP) of less than 1 and zero ozone depletion potential (ODP).
- Classified as A2L, they are mildly flammable and have low toxicity.
- These refrigerants are well-suited for use in scroll, screw, and centrifugal chillers, especially in commercial and institutional settings.
- In addition to pure HFOs, several blends have been introduced to serve as direct or near-drop-in replacements for legacy refrigerants:
    - R-513A is commonly used as a safer alternative to HFC-134a in large chillers, offering non-flammable performance with similar pressure characteristics[12].
    - R-444B is considered a candidate for retrofitting older R-22 systems in low-capacity chillers.
    - R-454B and R-452B are blends formulated to replace R-410A and R-407C respectively, combining lower GWP with operational efficiency[13].

While these refrigerants have seen increasing international deployment, uptake in India is still limited due to cost, availability, and current safety regulations. Nevertheless, their potential remains strong, especially as standards evolve and market acceptance improves.

b) Natural Refrigerants

Natural refrigerants are substances that occur in nature and have been used in refrigeration for over a century. In recent years, they are gaining prominence due to their zero ozone depleting potential and low global warming potential, cost-effectiveness, and suitability in specific applications.

- **Ammonia (R-717)** is widely used in industrial chillers due to its high efficiency and zero GWP. However, its **toxicity** and corrosive nature limit its application in densely populated buildings, despite its mature usage in cold chains and process cooling.
- **Propane (R-290)** has emerged as an effective option for small and medium-capacity chillers. It offers excellent thermodynamic performance but is classified as **highly flammable (A3),** which restricts its application in certain urban and commercial settings.
- **Carbon dioxide (R-744),** though non-toxic and non-flammable, operates under much higher pressures than traditional refrigerants. Its use in chillers is still emerging and often limited to **subcritical or cascade systems** in high ambient conditions.
- **Water (R-718),** used in **absorption and ejector-based chillers,** is both environmentally benign and safe but has limited cooling capacity and is often suitable only in **specialized applications** where waste heat is available.

Each of these refrigerants is already in use in India to varying extents such as ammonia in industrial units, propane in pilot commercial applications, and water-based systems in niche setups. Their future adoption will depend on advances in safety protocols and component availability.

c) Next-Generation HFCs

HFC-32 offer a transitional solution as the market gradually shifts toward low-GWP options.

- HFC-32, with a GWP of 675, is now widely used in split systems and commercial scroll chillers. It has higher efficiency than R-410A and requires a smaller refrigerant charge, which makes it well-suited for compact systems.
- It is classified as A2L, with mild flammability, and is increasingly supported by Indian manufacturers.
- HFC-32 also serves as a base component in several newer blends:
  - R-454B, developed as a lower-GWP substitute for R-410A, offers improved performance and reduced emissions.
  - R-452B is another A2L blend with properties suited for replacing R-407C in existing systems.
  - These refrigerants represent a practical solution for systems that are not ready for natural or HFO-based refrigerants due to design constraints or economic limitations [14].

Table 2.1, Comparison of low-GWP refrigerants by type, GWP, safety class, and adoption in India.

Refrigerant	Type	GWP	Flammability Class	Toxicity	Maturity in India	Application Suitability
R-1234yf/ze	HFO	<1	A2L (mild)	Low	Limited pilots	Medium to large chillers
R-290	HC	3	A3 (high)	Low	Pilot-scale use	Small/medium-sized chillers
R-717 (Ammonia)	Natural	0	B2L (mild)	High	Industrial use mature	Industrial, large buildings
CO_2 (R-744)	Natural	1	A1 (none)	Low	Experimental	Booster/transcritical systems
R-32	HFC	675	A2L (mild)	Low	Rapid adoption	Commercial scroll chillers
Water (R-718)	Natural	0	A1 (none)	None	Niche adoption	Absorption/ejector chillers
R-444B	HFO Blend	~296	A2L (mild)	Low	Emerging	R-22 retrofit option for small chillers
R-513A	HFO Blend	~631	A1 (non-flammable)	Low	Under limited use/testing	HFC-134a retrofits in screw/centrifugal
R-454B	HFO Blend	~466	A2L (mild)	Low	OEM trials ongoing	R-410A replacement in packaged chillers
R-452B	HFO Blend	~676	A2L (mild)	Low	Very limited pilots	R-407C replacement in retrofitted systems

ii. Comparative Technical Assessment of Low-GWP Refrigerants

The selection of refrigerants plays a critical role in determining the environmental performance, safety, and operational efficiency of chiller systems. India is transitioning from high-GWP substances under the Kigali Amendment, a variety of non-ODS and low-GWP alternatives are now available. This section compares these refrigerants across key technical parameters, incorporating both pure and blended options currently gaining traction in the chiller market.

a) Classification of Refrigerants

Refrigerants are commonly assessed based on two parameters: Global Warming Potential (GWP) and Ozone Depletion Potential (ODP). The refrigerants considered here are all non-ozone-depleting (ODP = 0), but differ significantly in their GWP values (Tables 2.2 and 2.3).

Table 2.2, Summary of Key Refrigerants

Refrigerant	Type	GWP (100-year)	Notes
R-1234yf / R-1234ze	HFO	<1	Ultra-low GWP; close thermodynamic match to R-134a
R-290 (Propane)	HC (Natural)	3	Excellent efficiency; flammable
R-717 (Ammonia)	Natural	0	High energy performance; toxic
R-744 (CO_2)	Natural	1	Zero flammability; high-pressure systems
R-32	HFC	675	Moderate GWP; widely used in modern systems
R-718 (Water)	Natural	0	Only applicable in absorption chillers
R-444B	HFO Blend	~296	Low-GWP alternative to R-410A; A2L safety classification
R-513A	HFO Blend	~631	A1 safety class; suitable as drop-in for R-134a
R-454B	HFO Blend	~466	Energy-efficient and climate-friendly option
R-452B	HFO Blend	~676	Performance close to R-410A with lower discharge temperature

b) Safety Classification

Using ASHRAE-34 safety classes, refrigerants are designated by toxicity (A = low, B = high) and flammability (1 = none, 2L = mild, 2 = moderate, 3 = high).

- **A1 (Non-flammable, Low Toxicity):** R-513A, R-744, R-718
- **A2L (Mildly Flammable, Low Toxicity):** R-1234yf, R-1234ze, R-454B, R-444B, R-452B, R-32
- **A3 (Highly Flammable):** R-290
- **B2L (Toxic, Mild Flammability):** R-717 (Ammonia)[15]

Table 2.3, Thermodynamic and Safety Characteristics of Low GWP Refrigerants.

Refrigerant	ASHRAE Class	Flammability	Toxicity	Use Case Suitability
R-1234yf/ze	A2L	Mild	Low	Scroll/screw/centrifugal chillers
R-290	A3	High	Low	Small-scale systems with strict safety
R-717	B2L	Low	High	Industrial and utility chillers
R-744	A1	None	Low	Cascade systems; high pressure
R-32	A2L	Mild	Low	Widely used in room and small commercial AC
Water (R-718)	A1	None	None	Absorption chillers
R-444B	A2L	Mild	Low	Residential/light commercial systems
R-513A	A1	None	Low	Drop-in for R-134a in chillers
R-454B	A2L	Mild	Low	Alternative to R-410A, better GWP
R-452B	A2L	Mild	Low	Suitable for transport cooling and retrofits

c) Performance and Suitability

These refrigerants differ not only in climate impact but also in thermodynamic behaviour, safety, and system integration. Several are compatible with existing equipment, while others require specialized components. Table 2.4 is presenting the uses, major strengths and limitations of various refrigerant in chiller system.

A wide range of low-GWP refrigerants is now technically available for chiller applications in India. While natural refrigerants like R-717 and R-290 offer higher environmental performance, but having safety and regulatory challenges. On the other hand, HFOs and HFO blends, such as R-1234ze and R-454B, are being rapidly adopted globally and are well-suited for modern high-efficiency systems. R-32 continues to serve as a transitional refrigerant due to its moderate GWP and widespread availability. The choice of refrigerant should consider not only environmental goals but also building type, system scale, safety requirements, technician capacity, and long-term serviceability. These trade-offs will be further explored through case studies in the upcoming chapter of the report.

Table 2.4, Technical Complexity and Retrofit Feasibility

Refrigerant	Typical Use Case	Key Strengths	Major Limitations
R-1234yf/ze	Commercial chillers (new systems)	Low GWP, good efficiency	Mild flammability, higher cost
R-290	Small/medium chillers	Very high efficiency	Highly flammable (A3); usage restrictions
R-717	Industrial facilities	Excellent COP; widely used in cold chains	Toxicity, skilled handling required
R-744	Supermarkets, hospitals (CO_2 racks)	Non-flammable, sustainable	High pressure design; not ideal in hot climates
R-32	Scroll/inverter chillers	Balanced GWP, readily available	Mildly flammable; moderate GWP
R-718	Absorption cooling (waste heat)	Zero emissions, freely available	Limited to special systems; low COP
R-444B	Scroll systems	Lower GWP than R-410A	Still A2L; needs new safety protocols
R-513A	Screw and centrifugal chillers	A1 refrigerant; drop-in for R-134a	Moderate GWP; less widely available in India
R-454B	VRF and packaged systems	Lower GWP and better performance than R-410A	Mild flammability; OEM adoption still growing
R-452B	Transport and packaged chillers	Good capacity match with R-410A	GWP still relatively high for long term use

iii. Transition Pathways for Existing Chillers Using Low-GWP Refrigerants

As India prepares for a long-term shift toward environmentally sustainable cooling technologies, the transition of existing chiller systems many of which still rely on high-GWP refrigerants like HFC-134a, R-410A, and R-407C becomes a critical focus. This section outlines practical pathways for replacing high-GWP refrigerants with suitable low-GWP alternatives based on system type, refrigerant class, and retrofit compatibility (Tables 2.5 and 2.6)[16].

a) R-22 Systems (HCFC)

Chillers using R-22, though now discontinued for new installations, still account for a significant number of operational units, particularly in medium-sized commercial buildings.

- Suggested Alternatives: R-290, R-32, R-444B

- Transition Strategy:
  - R-444B and R-32 offer a transitional path with moderate system modifications, such as oil replacement and valve recalibration.
  - R-290, while highly efficient, requires strict safety compliance due to its A3 flammability rating and is best suited for isolated or low-charge systems.
- **When to Retrofit**: Retrofit may be considered if the unit is less than 10–12 years old and in good mechanical condition.
- **When to Replace**: Replacement is advisable for older systems or where flammability codes restrict retrofitting.

b) HFC-134a Systems

HFC-134a chillers are widely used in large centrifugal and screw systems in offices, airports, and hospitals. While safe and familiar, their high GWP necessitates a planned transition.

- Suggested Alternatives: R-513A, HFO-1234ze
- Transition Strategy:
  - R-513A is a near drop-in with minimal hardware changes, ideal for extending the life of existing chillers.
  - HFO-1234ze may require compressor changes and polyolester (POE) oil, but offers ultra-low GWP performance.
- **When to Retrofit**: Recommended for systems under 10 years with reliable compressors.
- **When to Replace:** Replace if the system is reaching end-of-life or requires major component replacements.

c) R-410A Systems

Scroll chillers using R-410A are common in modern buildings due to their compactness and performance. However, R-410A has a GWP of over 2,000, prompting the need for lower-GWP alternatives.

- **Suggested Alternatives**: **R-32**, **R-454B**
- **Transition Strategy**:
  - **R-32** offers improved efficiency and a GWP nearly one-third that of R-410A. Retrofit feasibility depends on system age and design, especially for inverter-driven units.
  - **R-454B** is a newer A2L blend with similar performance to R-410A but a much lower climate impact.
- **When to Retrofit**: Retrofit possible if the system is relatively new and manufacturer-approved conversion kits are available.
- **When to Replace**: Older systems (>10 years) or those with proprietary control logic may require full replacement.

d) R-407C Systems

Originally introduced as a transitional replacement for R-22, R-407C has found wide application in packaged and scroll-based chillers. However, it still has a relatively high GWP.

- **Suggested Alternatives**: **R-32**, **R-452B**
- **Transition Strategy**:
  - **R-452B** is engineered to closely match R-407C performance characteristics, allowing for controlled retrofitting with valve and oil adjustments.
  - **R-32** can also be considered, but requires evaluation of operating pressure and discharge temperature compatibility.
- **When to Retrofit**: Feasible in systems with accessible components and moderate runtime.
- **When to Replace**: Recommended if the system has frequent service needs or lacks OEM support for conversion.

Table 2.5, Summary Table: Transition Pathways for Common Refrigerants

Existing Refrigerant	Suggested Low-GWP Alternatives	Recommended Action	When to Retrofit	When to Replace
R-22	R-290, R-32, R-444B	Retrofit or replace (case-dependent)	Unit <12 years, good condition, low-charge applications	Older than 12 years, frequent leaks, safety restrictions
HFC-134a	R-513A, HFO-1234ze	Retrofit with system flushing/ oil change	System <10 years, reliable compressor	Aging system or major component failure
R-410A	R-32, R-454B	OEM retrofit or full replacement	OEM retrofit kits available, system <8–10 years	End-of-life systems or proprietary electronics
R-407C	R-32, R-452B	Retrofit with expansion valve adjustments	Recently serviced, moderate runtime	If system has performance issues or lacks conversion support

The transition of existing chillers to low-GWP refrigerants requires a balanced approach that considers system age, application type, safety classification, and technical compatibility. While retrofitting offers a cost-effective path in many cases, replacing outdated systems with new, energy-efficient chillers pre-charged with environmentally sustainable refrigerants is often the better long-term strategy. The refrigerant roadmap from High-GWP to Low-GWP is illustrated in Fig. 2.2 and Table 2.6. As awareness improves and safety standards evolve, India's readiness to adopt these alternatives will continue to grow—supported by OEM guidance, technician training, and clear regulatory direction [17].

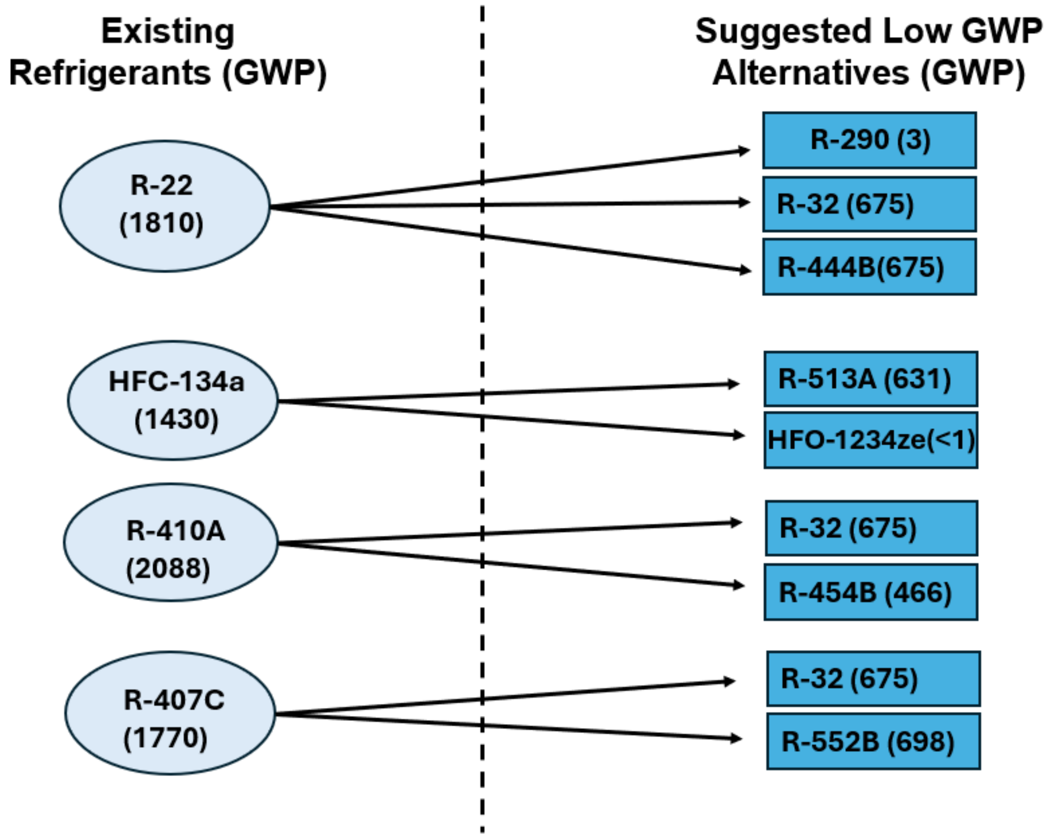


Fig. 2.2 Refrigerant roadmap from High-GWP to Low-GWP

Table 2.6, Chiller Types and Their Compatible Low-GWP Refrigerants

Chiller Type	Compatible Low-GWP Refrigerants	Cooling Capacity Range	Typical Applications	Adoption Status in India
Scroll Chillers	R-290, R-1234yf, R-32, R-454B, R-452B, R-444B	10 – 200 TR	Small commercial buildings, hotels, offices	Growing in metro areas and premium green buildings
Screw Chillers	R-717, R-1234ze, R-513A, R-290	100 – 800 TR	Hospitals, data centers, mixed-use buildings	Moderate adoption; legacy systems dominate
Centrifugal Chillers	R-1234ze, R-1233zd, R-514A, R-513A, R-718	>400 TR	Airports, malls, large institutions, district cooling	Increasing deployment in government/institutional buildings
Absorption Chillers	R-718 (water-LiBr), Ammonia-Water (R-717)	100 – 1000+ TR	Solar cooling, industries with waste heat, campuses	Established in energy-intensive government/ industrial setups
CO_2 Transcritical Systems	R-744	20 – 150 TR (modular)	Hospitals, retail cold chains, green campuses	Pilot stage in Indian hospitals and cold storage chains
Magnetic Bearing Chillers	R-1234ze, R-513A	200 – 1000+ TR	High-rise buildings, green-certified IT parks	Limited but emerging for premium buildings

2.1 Types of chillers and applications

India's transition to low-GWP refrigerants is supported by a diverse range of chiller technologies tailored for different building sizes, system capacities, and performance needs. As high-GWP refrigerants such as R-22, R-134a, and R-410A are gradually phased out, both global and domestic manufacturers have introduced chillers compatible with next-generation refrigerants. These include natural refrigerants, hydrofluoroolefins (HFOs), and several HFO-HFC blends designed to offer energy-efficient and environmentally responsible alternatives. The Table 2.7 below outlines the key chiller types currently in use or under consideration in India, along with the low-GWP refrigerants they support, their typical application areas, and the extent of their deployment[18].

i. Refrigerants suitable for different types of Chillers

- **Scroll chillers** are well suited for newer refrigerants such as R-32 and R-1234yf, and are commonly deployed in mid-size commercial buildings. R-452B and R-454B are now being introduced as replacements for R-410A, with some models also compatible with R-444B.
- **Screw chillers**, which serve larger buildings and continuous load applications, are compatible with a wide range of refrigerants including ammonia (R-717), propane (R-290), and non-flammable blends such as R-513A. These systems offer flexibility in both greenfield and retrofit applications.
- **Centrifugal chillers**, typically used in large infrastructure projects, are gradually moving toward HFOs like R-1234ze and blends such as R-513A. While these refrigerants require system modifications, they offer improved environmental performance without compromising cooling output.
- **Absorption chillers**, powered by waste heat or solar energy, continue to be a sustainable option in campuses and industrial clusters. These systems use water or ammonia as working fluids and contribute to both energy savings and emissions reduction.
- **CO_2-based systems (R-744)**, though more common in refrigeration, are being tested for high-density urban cooling applications. Their potential for heat recovery and compact design makes them suitable for specialized installations.
- **Magnetic bearing chillers**, known for oil-free operation and high efficiency, are compatible with low-pressure refrigerants like R-1234ze. Though still relatively expensive, they are increasingly considered in projects aiming for LEED or GRIHA certification.

ii. Existing Refrigerants Currently Used in Chillers

Chillers remain the backbone of centralized cooling systems in large commercial, institutional, and industrial buildings across India. Despite the availability of emerging low-GWP options, the installed base of chillers in the country still predominantly relies on conventional refrigerants—mainly HCFCs and HFCs (Fig. 2.3), that were adopted during the past two decades due to their availability, performance characteristics, and established service infrastructure. This section outlines the most commonly used refrigerants in existing chiller systems, their applications, thermodynamic properties, and limitations (Table 2.7 and Fig. 2.3).

a) HCFC-22 (R-22)

R-22, a hydrochlorofluorocarbon (HCFC), was widely adopted in air-conditioning and chiller systems due to its stable thermodynamic properties, moderate pressure requirements, and proven energy performance.

- **GWP**: ~1,810
- **ODP**: 0.055
- **Applications**: Reciprocating, scroll, and screw chillers in mid- to large-scale commercial buildings
- **Current Use**: Legacy systems still operational in hospitals, malls, and institutional campuses, though new installations are no longer permitted
- **Limitations**: Significant ozone depletion potential, high GWP, limited future availability, and increasingly expensive servicing

R-22 is phased-out under India's HCFC Phase-Out Management Plan (HPMP), a substantial number of units remain in service. These are primarily maintained through reclaimed or stockpiled refrigerant, raising long-term sustainability and maintenance concerns.

b) HFC-134a

HFC-134a, a hydrofluorocarbon with zero ODP, has been the dominant refrigerant in medium to large water-cooled and air-cooled chillers since the early 2000s.

- **GWP**: ~1,430
- **ODP: 0**
- **Applications**: Centrifugal and screw chillers used in office towers, airports, hospitals, and convention centers
- **Performance**: High COP in moderate ambient conditions, low compressor discharge temperature
- **Status**: Still widely used in new equipment and aftermarket servicing across India

While HFC-134a does not damage the ozone layer, its high GWP contributes to greenhouse gas emissions. It remains common due to its safety classification (A1 – non-flammable, non-toxic), compatibility with existing system designs, and extensive servicing expertise available in the market.

c) HFC-410A

A zeotropic blend of R-32 and R-125, HFC-410A became popular in commercial air-conditioning and packaged chiller systems as a replacement for R-22.

- **GWP**: ~2,088
- **ODP**: 0
- **Applications**: Scroll and rotary chillers used in hotels, institutional buildings, and mid-size offices
- **Advantages**: Good energy efficiency, moderate operating pressures
- **Limitations**: High GWP, limited retrofit potential due to pressure and lubricant incompatibility

Although effective in terms of performance, HFC-410A faces regulatory pressure globally. Several countries have already restricted its use in new equipment under their national HFC phase-down plans.

d) HFC-407C

HFC-407C is a blend refrigerant designed to serve as a retrofit option for R-22 systems and a transitional solution for moderate-temperature air-conditioning systems.

- **GWP**: ~1,770
- **Applications**: Scroll chillers, retrofitted R-22 units in commercial buildings
- **Properties**: Exhibits glide during phase change, requiring precise system design
- **Challenges**: Lower efficiency in high ambient conditions; phase glide may affect heat exchanger performance

While not as widely adopted as HFC-134a or HFC-410A in new chillers, HFC-407C continues to be used in retrofits, especially where conversion from R-22 is being attempted without full system replacement.

Overall, the current chiller fleet in India largely reflects the global transition trajectory that preceded Kigali Amendment enforcement. Most chillers still rely on refrigerants that, although ozone-safe, have high global warming potential. The predominance of HFC-134a and R-410A in commercial installations and the legacy use of R-22 underscore the urgency of coordinated policy and market action to facilitate a sustainable transition. The ongoing phase-down strategy outlined in HPMP Stage-III, combined with India's proposed HFC phase-down schedule under the Kigali framework, is expected to significantly alter refrigerant use patterns over the next decade. However, widespread change will depend on factors such as:

- Availability of retrofit-friendly refrigerants
- Cost of system upgrades
- Awareness among facility owners and HVAC contractors
- Institutional mechanisms to support the adoption of low-GWP alternatives

Table 2.7, Existing Refrigerants Currently Used in Chillers

Refrigerant	Type	GWP	Common Applications	Key Characteristics	Current Use in India	Remarks
R-22	HCFC	1810	Legacy screw/scroll chillers in hospitals, malls, institutions	Good thermodynamic performance; ODP = 0.055; being phased out under HPMP	Still present in older systems; new use banned post-2024	Phased out under HPMP; Used for servicing of old systems
R-134a	HFC	1430	Centrifugal and screw chillers in airports, offices, hospitals	A1 safety class; widely used; zero ODP; high GWP	Dominant in installed base; still used in new chillers	Still widely used; high GWP but non-ozone depleting
R-410A	HFC (Blend)	2088	Scroll chillers in hotels, campuses, commercial buildings	Efficient; higher pressure; not retrofit-friendly	Used in many mid-size buildings	Popular R-22 alternative; high pressure and limited retrofit potential
R-407C	HFC (Blend)	1770	Packaged Scroll chillers and retrofitted R-22 systems	Zeotropic blend; phase glide; moderate performance	Used in retrofits and select commercial applications	Used in R-22 retrofits; reduced performance at high ambient temperatures

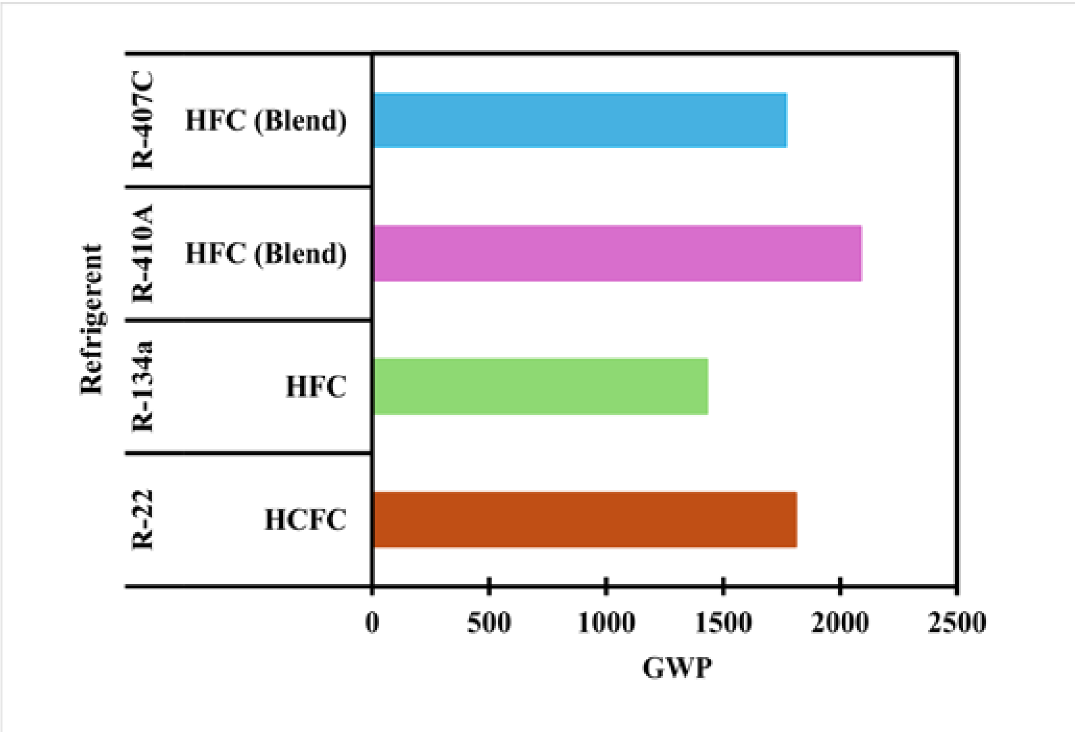


Fig. 2.3 Global warming potential (GWP) of common refrigerants.

iii. Additional Refrigerants and Their Roles

Recent refrigerant blends such as R-444B, R-513A, R-454B, and R-452B have been introduced to serve as direct replacements for high-GWP refrigerants in both new and retrofit applications.

- R-444B offers a lower GWP option to replace R-410A and is compatible with scroll systems.
- R-513A is a drop-in substitute for R-134a, offering comparable performance with improved climate impact, particularly in screw and centrifugal chillers.
- R-454B and R-452B are A2L-class refrigerants gaining acceptance as improved alternatives to R-410A, with better efficiency and reduced flammability risks.
- These refrigerants offer flexible pathways for transitioning both existing and new systems, provided that appropriate safety measures and design adjustments are observed.

India's commercial and institutional cooling landscape is well-positioned to benefit from the adoption of low-GWP refrigerants. With multiple chiller types supporting a growing range of climate-friendly refrigerants, the choice of technology must be guided by building size, system load profile, safety regulations, and long-term operating costs. The increasing availability of refrigerant blends for retrofit and new installations, combined with advancements in chiller design, provides a timely opportunity for market transformation. Going forward, updated safety codes, technician training, and greater supply chain access will be essential to accelerate this transition across building sectors.

iv. Evaluation of Key Chiller Performance Metrics

The performance of low-GWP refrigerants in chiller systems is a function of their thermodynamic properties, energy efficiency, safety classification, and operational reliability. This section evaluates and compares the key performance metrics of selected refrigerants relevant for the Indian commercial air-conditioning market, focusing on Coefficient of Performance (COP), Energy Efficiency Ratio (EER), Global Warming Potential (GWP), and ASHRAE Safety Classification. Additional considerations include refrigerant charge size, system cost, and maintenance complexity (Table 2.8)[19].

Table 2.8, Evaluation of Key Chiller Performance Metrics

Refrigerant	Type	COP Range (@ ARI* conditions)	EER Range	GWP (AR5)	ASHRAE Safety Class	Remarks
R-1234yf	HFO	4.5 – 5.5	15.35 – 18.77	<1	A2L	Mildly flammable, similar to R-134a in performance
R-1234ze(E)	HFO	5.0 – 6.2	17.06 – 21.15	<1	A2L	High efficiency; suitable for centrifugal chillers
R-290 (Propane)	Hydrocarbon	4.7 – 6.0	16.04 – 20.47	3	A3	Excellent thermodynamic properties; A3 safety limits usage
R-717 (Ammonia)	Natural	5.5 – 7.0	18.77 – 23.88	0	B2L	Highest COP; toxic; used in large industrial plants
R-744 (CO_2)	Natural	3.2 – 4.0 (transcritical)	10.92 – 13.65	1	A1	High operating pressure; suitable for subcritical/ cascade systems
R-32	HFC	4.5 – 5.5	15.35 – 18.77	675	A2L	Moderate GWP; used in scroll chillers; mildly flammable
R-718 (Water)	Natural	0.7 – 1.0	2.39 – 3.41	0	A1	Used in absorption/ ejector chillers; non-compression-based
R-444B	HFO Blend (R-32/R-1234yf/R-152a)	4.2 – 4.8	14.32 – 16.38	~296	A2L	Lower GWP alternative to R-410A; mildly flammable
R-513A	HFO Blend (R-134a/R-1234yf)	4.3 – 5.0	14.67 – 17.06	~631	A1	Non-flammable; drop-in for R-134a applications
R-454B	HFO Blend (R-32/R-1234yf)	4.6 – 5.4	15.70 – 18.43	~466	A2L	Intended as R-410A replacement; lower flammability than R-32
R-452B	HFO Blend (R-32/R-125/R-1234yf)	4.5 – 5.2	15.35 – 17.74	~698	A2L	Designed for higher ambient performance; similar to R-410A

*ARI – Air-Conditioning and Refrigeration Institute

To support the evaluation of low-GWP refrigerants, graphical analyses was conducted—plotting Coefficient of Performance (COP) and Energy Efficiency Ratio (EER) against Global Warming Potential (GWP) (Figs. 2.4). These visualizations provide a clear comparative perspective on how different refrigerants balance thermal performance with environmental impact. The COP vs GWP analysis indicates that refrigerants such as R-717 (Ammonia) and R-290 (Propane) offer superior energy efficiency, with COP values exceeding 6.0 in favorable conditions, while maintaining GWP values close to zero. R-1234ze(E)[20], an HFO refrigerant, also demonstrates high efficiency with negligible climate impact, making it a strong candidate for centrifugal and large-scale chillers. In contrast, R-744 (CO_2), despite its extremely low GWP, shows a comparatively lower COP due to its performance limitations under transcritical operation, especially in high ambient temperature zones. Meanwhile, refrigerant blends such as R-452B, R-513A, and R-454B offer intermediate COP values in the range of 4.5 to 5.0 but are associated with higher GWP levels between 450 and 700 (Fig. 2.4). These trade-offs must be weighed carefully when selecting refrigerants for applications with strict safety or retrofit constraints.

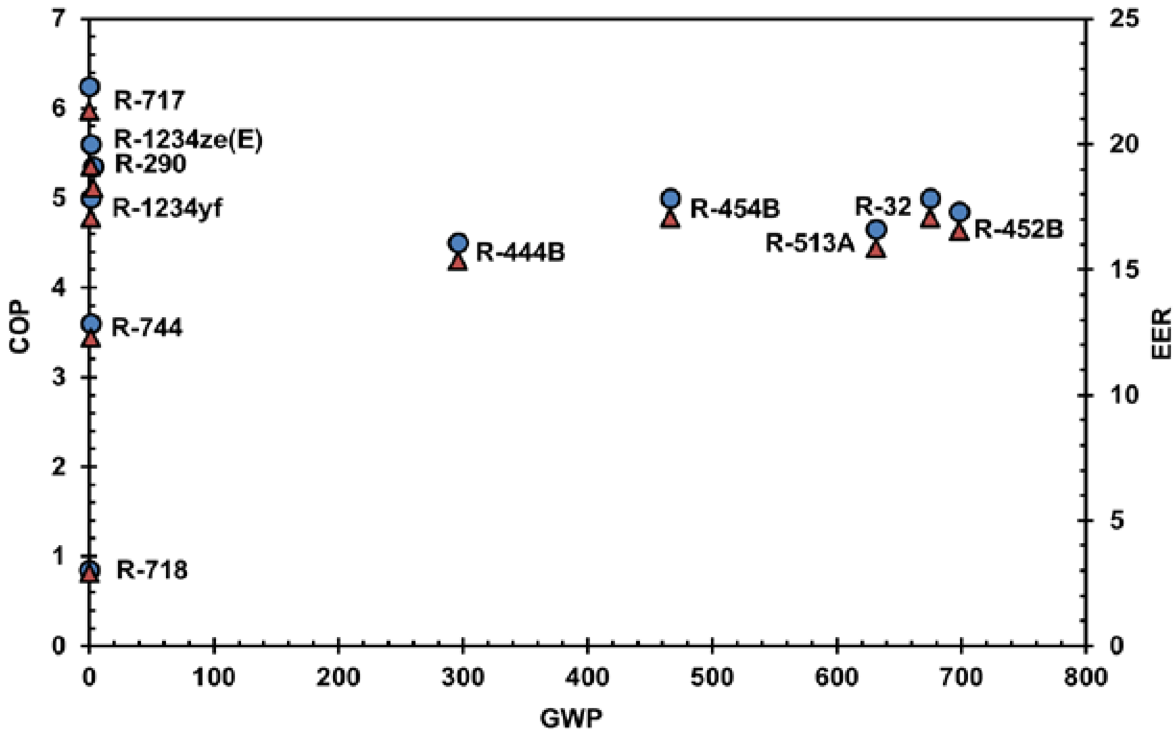


Fig. 2.4, COP and EER of selected refrigerants plotted against their GWP

When examining EER against GWP, a similar trend emerges. Natural refrigerants like Ammonia and Propane consistently deliver EERs above 20 BTU/hr·W, reinforcing their strong thermodynamic performance. R-1234yf and R-1234ze(E) maintain moderate to high EERs, ranging from 15 to 18 (Fig. 2.4), while offering very low GWP, making them viable for new system installations that prioritize climate mitigation.

Refrigerants such as R-452B and R-32 demonstrate competitive EER values but have significantly higher GWP, suggesting their use should be targeted to applications with limited flammability tolerance or in regions with less regulatory pressure. Water (R-718), although it has a GWP of zero, exhibits a much lower EER since it is primarily used in absorption-based systems that rely on thermal energy inputs rather than electric compressors.

Overall, the results from graphical analyses indicates that certain refrigerants, particularly Ammonia, Propane, and R-1234ze(E) and offer a favourable combination of high energy efficiency and minimal environmental burden (Fig. 2.4). However, their adoption at scale depends on multiple factors, including building type, safety code compliance, technician training, and compatibility with existing infrastructure.

2.2 Existing Technologies Used in Chillers

Sustainable cooling systems requires exploring not just alternative refrigerants, but also a broader range of energy-efficient and low-emission technologies (Fig. 2.5). As large air-conditioned buildings continue to drive peak electricity demand, the integration of advanced chiller systems offers a strategic pathway to reduce environmental impact and enhance system performance.

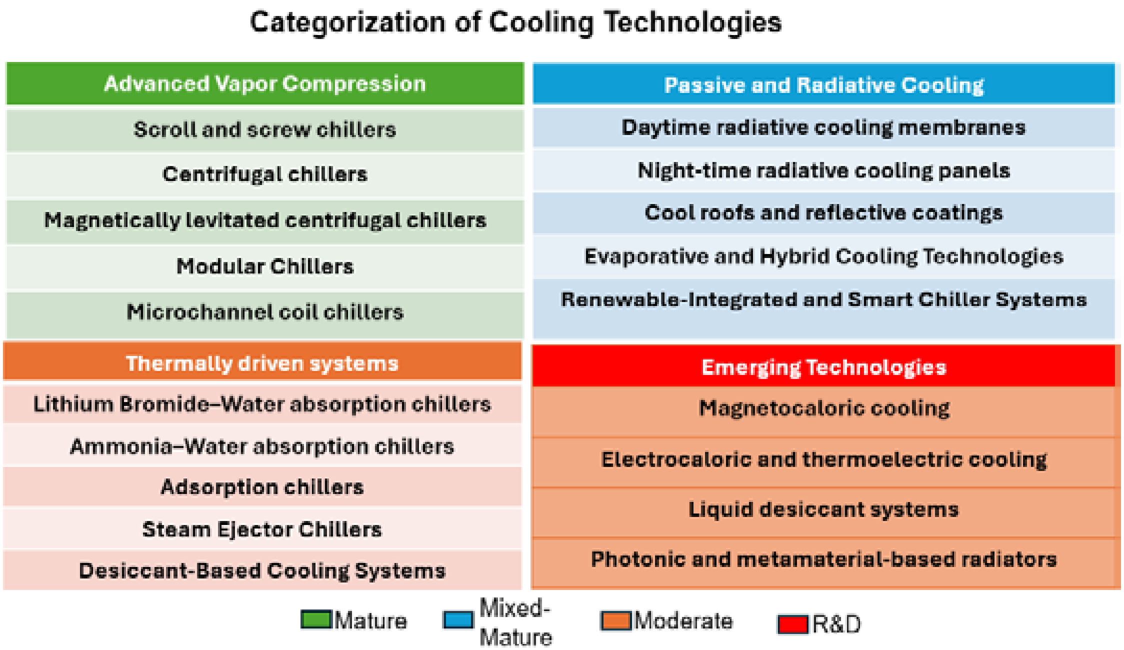


Fig. 2.5, Spectrum of cooling solutions for India.

i. Advanced Vapor Compression Chillers Using Low-GWP Refrigerants

Conventional vapor compression systems remain dominant in commercial buildings. However, significant improvements have been achieved through the adoption of low-GWP refrigerants, magnetic bearing compressors, inverter drives, and microchannel heat exchangers (Fig. 2.6). These systems offer high part-load efficiency and better environmental performance (Table 2.9).

Examples include:

- **Scroll and screw chillers** using R-290 (propane), R-32, and R-1234yf.
- **Centrifugal chillers** operating with ultra-low GWP refrigerants like R-1234ze and R-514A.
- **Magnetic levitation compressors**, now available in Indian airports and hospitals, offer oil-free operation and reduced maintenance.

These technologies are commercially viable and can be deployed immediately in both new and retrofitted infrastructures.

Table 2.9, Overview of Advanced Low-GWP Vapor Compression Chiller Technologies

Chiller Type / Technology	Key Features / Description	Low-GWP Refrigerants Used	GWP Range	Typical Use Cases
Magnetically levitated centrifugal chillers	Oil-free operation, high part-load efficiency, reduced maintenance	R-1234ze, R-513A, R-134a	< 10	Airports, data centers, hospitals
Inverter-driven scroll/screw chillers	Variable-speed control, high part-load performance	R-290, R-32, R-1234yf	< 675	Hotels, commercial offices, malls
Microchannel coil chillers	Compact footprint, low refrigerant charge requirement	R-1234ze, R-717	0–6	ECBC+ buildings, retrofits
Scroll Chillers	Suitable for small–medium cooling loads	R-290, R-32, R-1234yf	< 675	Commercial buildings, modular units
Screw Chillers	Efficient for continuous medium–large loads	R-1234ze, R-717 (ammonia)	0–6	Institutions, industrial premises
Centrifugal Chillers	Central plant applications, high-capacity	R-1233zd(E), R-514A	< 2	Airports, district cooling, large campuses
Modular Chillers	Easily scalable systems, suitable for phased expansion	R-32, R-290	< 675	Greenfield buildings, SEZs, smart cities

a) Magnetically levitated centrifugal chiller

b) Inverter-Driven Scroll Chiller

c) **Air-cooled Magnetic Centrifugal Chiller**

d) **Air Cooled DC Inverter Scroll Chillers**

Fig. 2.6, Advanced Vapor Compression Chillers Capable of Using Low-GWP Refrigerants

ii. Thermally Driven Chillers

Thermally driven systems operate without compressors and rely on heat sources solar thermal, industrial waste heat, or hot water to drive the cooling process (Fig. 2.7). These systems do not use conventional synthetic refrigerants and are particularly relevant in settings with consistent heat availability (Table 2.10).

Key technologies include:

- **Lithium Bromide–Water absorption chillers**, Use water as refrigerant and LiBr as absorbent, driven by hot water/steam; widely used in hospitals and district cooling; simple but limited to moderate cooling.

Ammonia–Water absorption systems, Use ammonia–water pair, suitable for industrial-scale cooling; efficient with zero GWP but safety concerns due to ammonia toxicity.

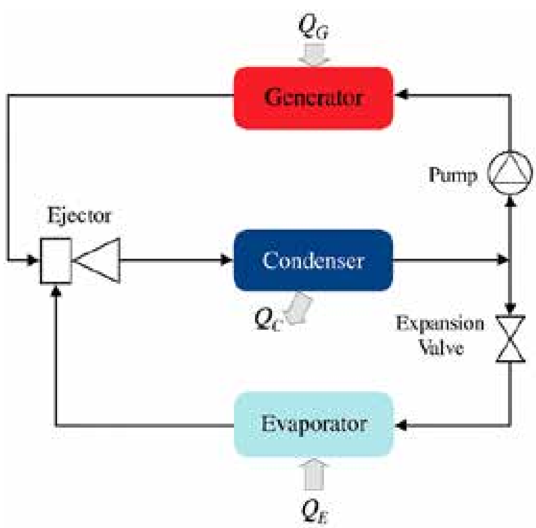
- **Adsorption chillers** Employ silica gel/zeolite with low-grade heat (solar/waste); eco-friendly, simple, but bulky and low efficiency.
- **Steam ejector systems**, Use steam jets for cooling; very simple and low-cost, best for industries with surplus steam; efficiency is poor.

These systems are proven and have been deployed in various institutional and industrial settings across India.

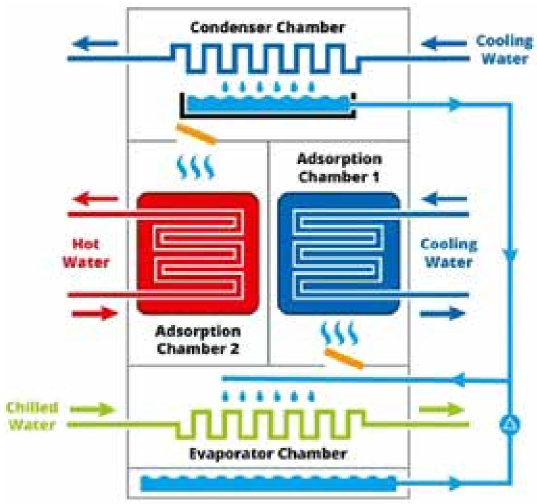
Table 2.10, Overview of Thermally Driven and Desiccant-Based Cooling Technologies

System Type	Technology / Working Principle	Typical COP	Key Features / Remarks
LiBr–Water Absorption Chiller	Heat-driven desorption of LiBr solution (water as refrigerant)	0.7–1.1	No compressor; widely adopted; ideal for solar/waste heat
Ammonia–Water Absorption Chiller	Uses ammonia (refrigerant) and water (absorbent); high-temp driven	0.5–0.8	Zero GWP; suited for industrial applications
Adsorption Chillers	Solid desiccants (silica gel, zeolite) adsorb/desorb water vapor	0.4–0.7	Lower COP; works with low-grade heat sources (55–90°C)
Steam Ejector Chillers Cooling via expansion of steam in an ejector nozzle		0.1–0.3	Mechanically simple; used in industries with surplus steam
Desiccant-Based Cooling Systems	Removes latent heat using solid/liquid desiccants	—	Effective humidity control; suitable for hybrid and solar cooling

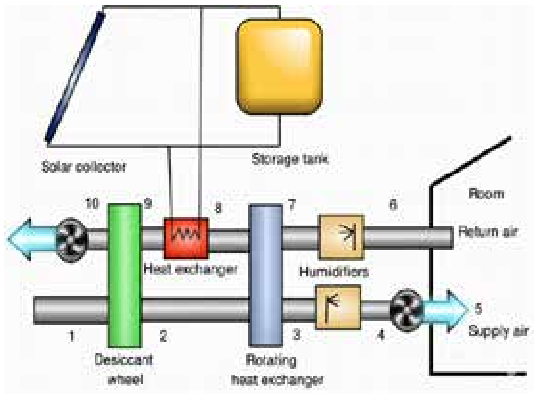
a) LiBr–Water Absorption Chiller



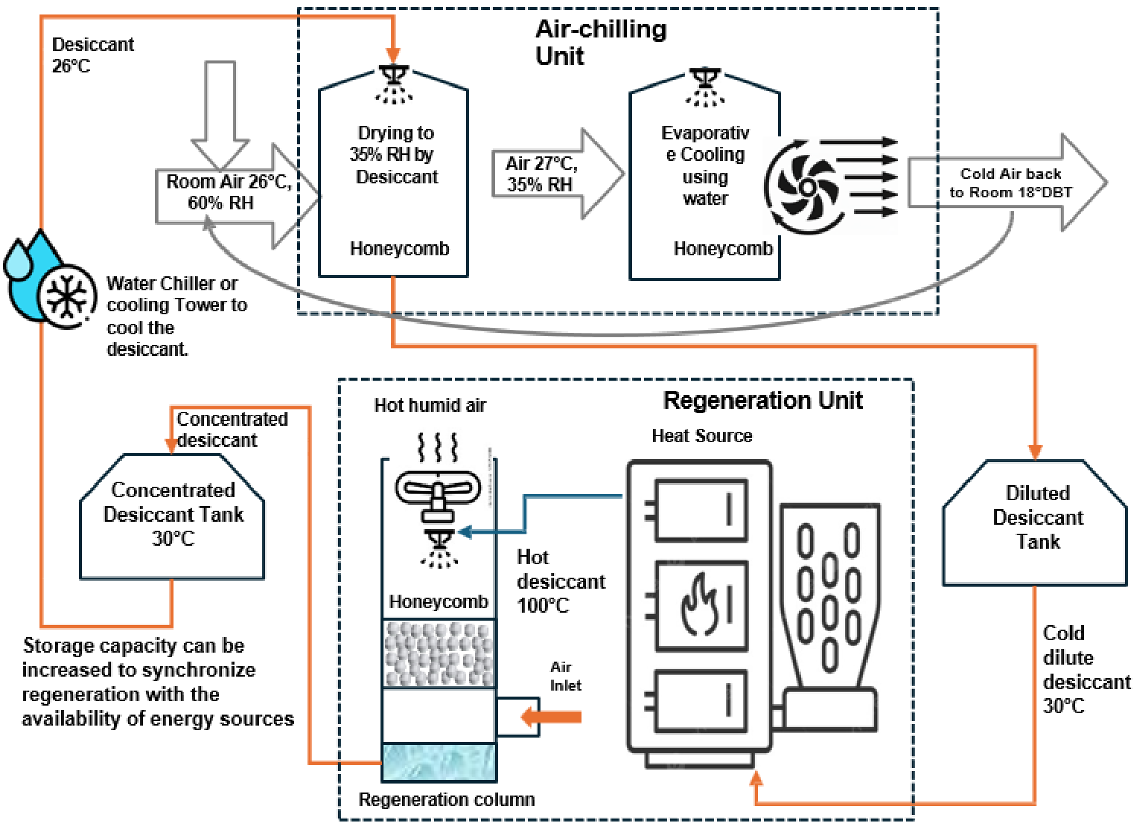
b) Ejector cooling system



c) Adsorption Chiller



d) Solid desiccant Cooling System



e) Liquid desiccant Cooling System

Fig. 2.7, Thermally Driven and Desiccant-Based Cooling Technologies

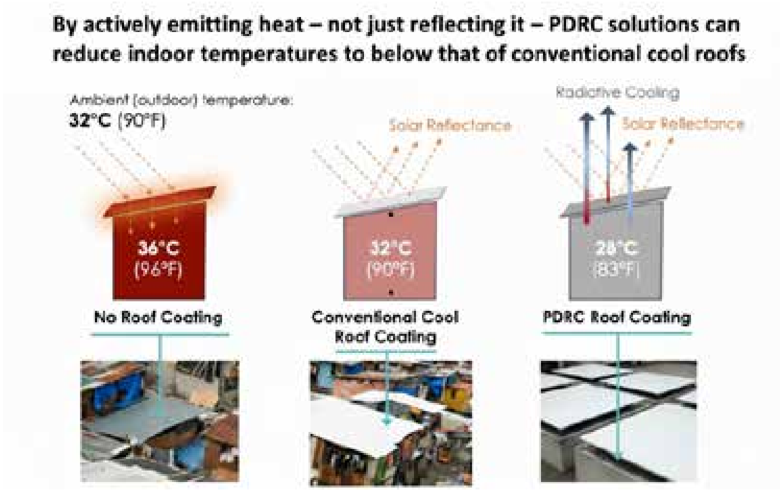
iii. Passive and Radiative Cooling Solutions

Passive cooling technologies eliminate or reduce the need for active refrigeration. By using natural processes such as radiative heat loss and reflective insulation, these systems offer a low-cost, low-energy alternative (Fig. 2.8 and Table 2.11).

Examples include:

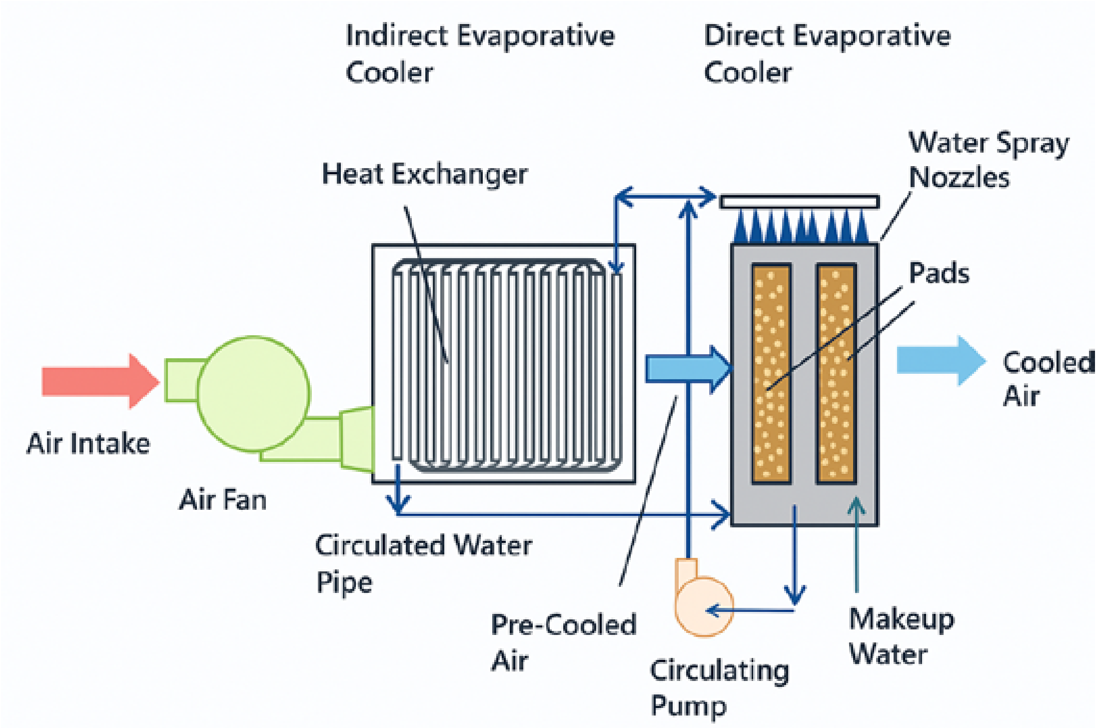
- **Daytime radiative cooling membranes**, which emit heat into the sky through the atmospheric transparency window.
- **Night-time radiative cooling panels**, suitable for regions with clear skies and dry climates.
- **Cool roofs and reflective coatings**, already promoted in urban housing schemes to reduce the cooling load of buildings.

These technologies are particularly well-suited to hot-arid and composite climate zones in India and require minimal maintenance.

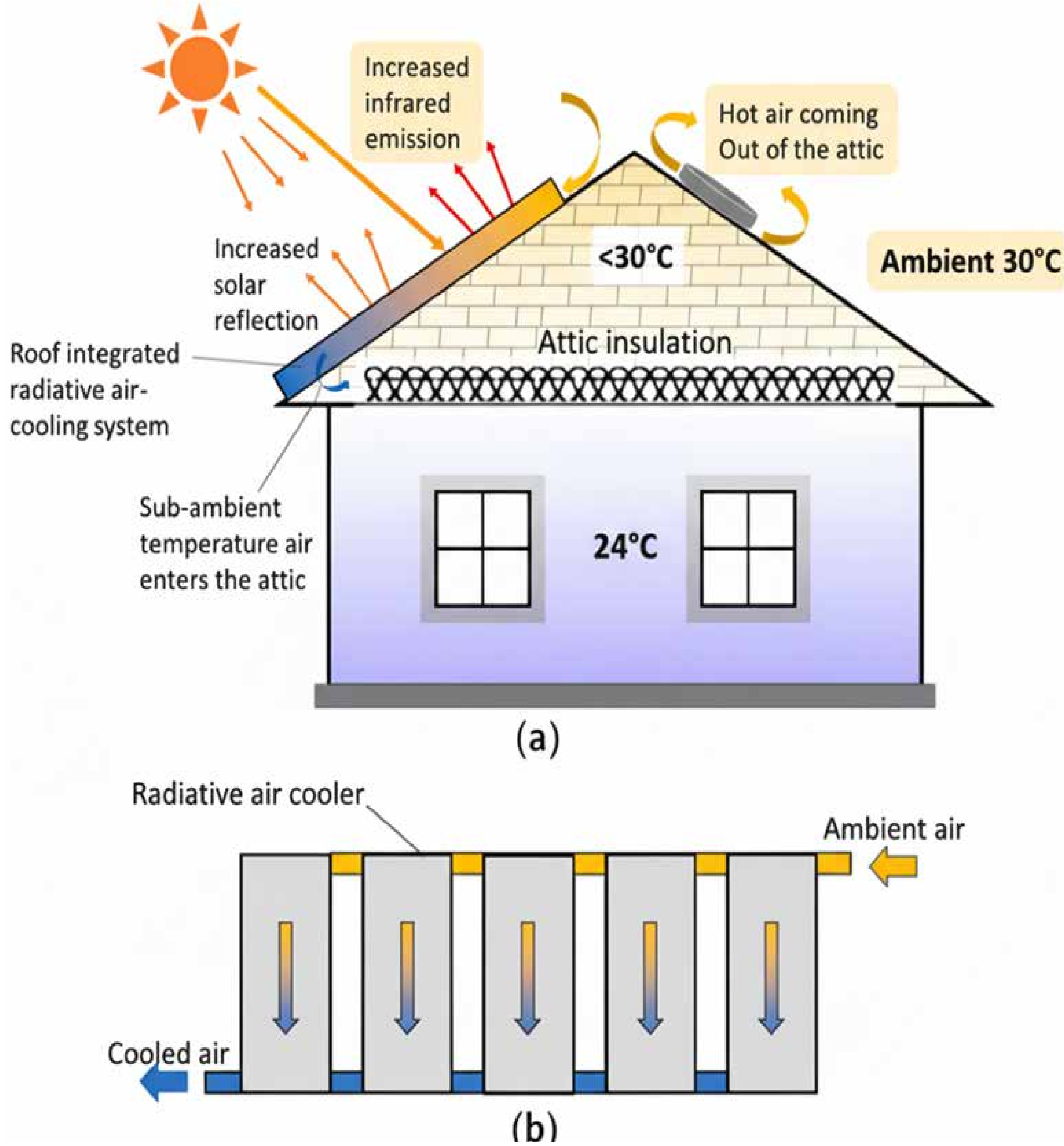


a) **Passive daytime radiative cooling**

b) **Chilled water thermal storage**



c) **Indirect and Direct Evaporative Cooling**



c) **Radiative cooling**

Fig. 2.8, Radiative and Evaporative Cooling Solutions

Table 2.11, Thermally Driven and Desiccant-Based Cooling Technologies Characteristics

Technology Type	Specific Technology	Mechanism	Application	Remarks/ Limitations
Radiative Cooling	Daytime Radiative Cooling (DRC)	Emits thermal radiation to sky window (8–13 μm)	Rooftops, facades, passive pre-cooling	Weather-dependent; limited performance in humid/cloudy areas
Radiative Cooling	Nighttime Radiative Panels	Surface heat loss to cold night sky	Rural and semi-urban applications	Cooling only during nighttime
Envelope Modification	Cool Roofs and Reflective Coatings	High-albedo surfaces reduce solar heat gain	Urban retrofits, government housing	Effectiveness depends on color durability and dust accumulation
Absorption Cooling	LiBr–Water or NH_3–Water Chillers	Heat-driven desorption cycles without compressors	District cooling, hospitals, industrial	Requires consistent heat input; bulky installations
Adsorption Cooling	Silica Gel–Water, Zeolite–Water	Solid sorbents desorb moisture at low temps	Buildings with low-grade waste heat	Lower COP; higher space requirements
Ejector Cooling	Steam Jet or Gas Ejector Systems	Uses momentum of motive fluid for entrainment	Industrial plants with surplus steam	Low COP; suitable only where steam is freely available
Desiccant Cooling	Solid or Liquid Desiccant Systems	Removes latent heat before sensible cooling	High humidity zones, hybrid HVAC	Needs periodic regeneration; works best when solar heat is available

iv. Evaporative and Hybrid Cooling Technologies

Evaporative systems use the natural cooling effect of water evaporation to lower air temperature. In dry climates, these systems offer high efficiency and significant energy savings (Table 2.12).

Variants include:

- **Direct evaporative cooling**, where air is cooled by direct water contact.
- **Indirect evaporative systems**, which cool air without adding humidity.
- **Hybrid systems** that integrate desiccant dehumidification and indirect cooling stages to provide both sensible and latent load control.

Such systems have been successfully deployed in large warehouses, data centers, and semi-industrial facilities in India.

Table 2.12, Evaporative and Hybrid Cooling Systems

System	Sub-Type	Efficiency (EER)	Best Use
Direct Evaporative Cooling	Air cooled directly via water contact	Up to 20+	Dry climates
Indirect Evaporative Cooling	Heat exchanger-based	10–14	Data centers, modular units
Two-stage IEC-DEC Systems	Combine both modes for higher cooling	>25	Industrial sheds, airports
Hybrid ACs with Desiccant Wheels	Latent + sensible cooling	Up to 15	Coastal/humid cities like Mumbai, Chennai

v. Renewable-Integrated and Smart Chiller Systems

These systems are designed for future-ready infrastructure, combining conventional or thermally driven chillers with renewable energy sources and thermal storage (Table 2.13 and Fig. 2.9).

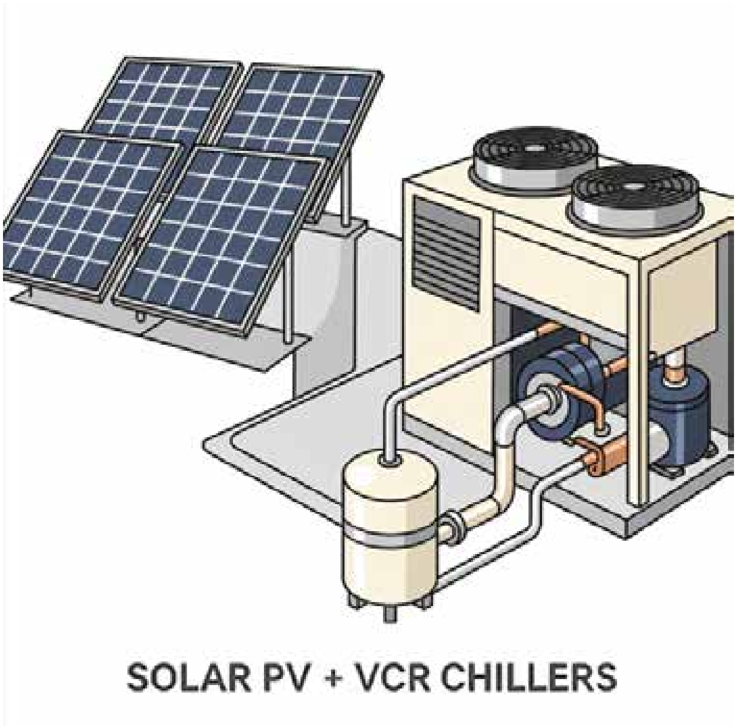
Prominent approaches:

- **Solar photovoltaic-assisted chillers:** Chillers powered directly by electricity generated from solar PV during daytime operation, reducing dependence on grid electricity and lowering carbon footprint (Fig. 2.9).
- **Solar thermal-driven absorption chillers:** Use solar thermal collectors to provide the heat input for LiBr–water or ammonia–water absorption systems, successfully demonstrated in regions with high solar insolation (Fig. 2.9).
- **Thermal energy storage systems:** Store cooling energy in chilled water tanks or phase change materials (PCMs) during off-peak hours, allowing peak load shifting, grid balancing, and improved system efficiency (Fig. 2.9).
- **Smart chillers with IoT Controls:** Integrated with sensors and digital platforms for real-time monitoring, predictive maintenance, and demand response, ensuring optimized performance and energy savings.

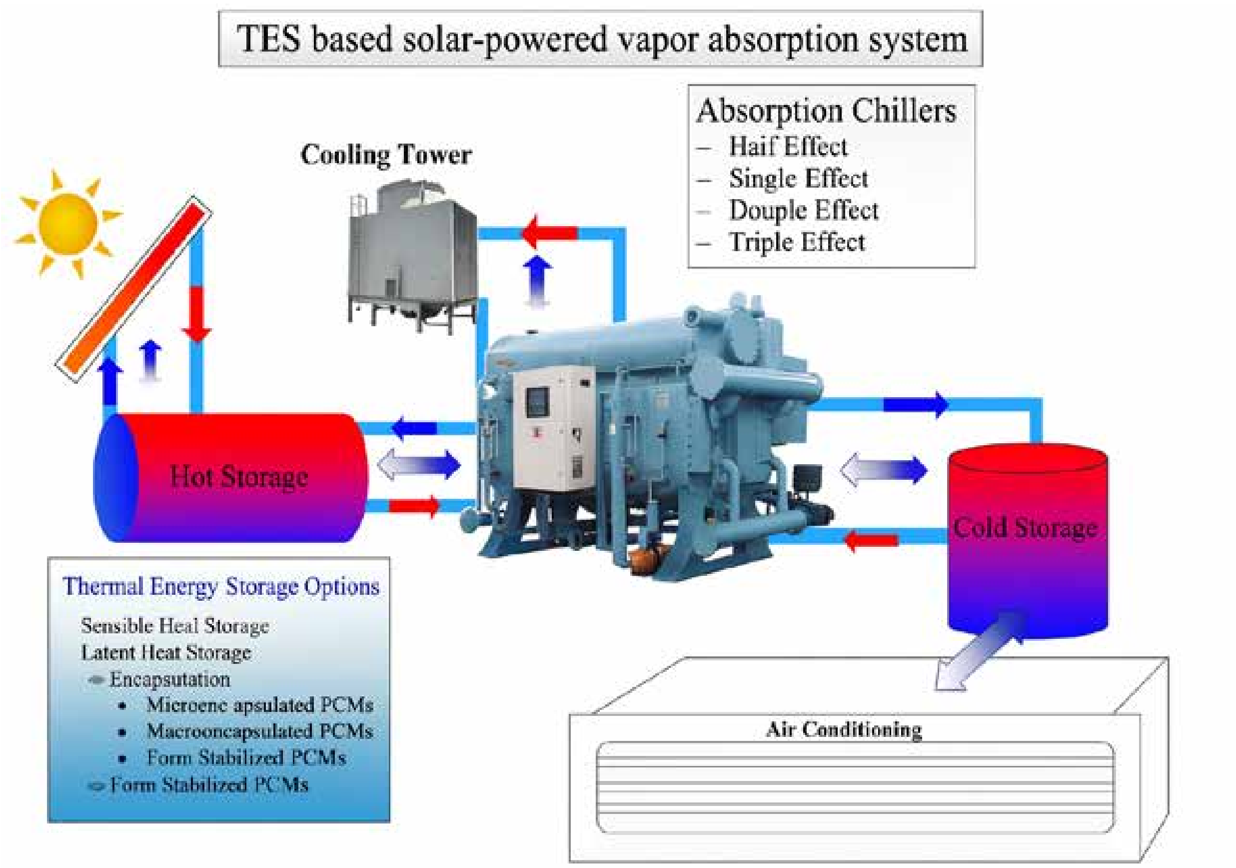
These systems are especially relevant for smart cities, SEZs, and mission-driven public buildings seeking to achieve net-zero operational targets.

Table 2.13, Renewable-Integrated and Smart Cooling Systems

System	Description / Key Feature	Integration / Benefits
Solar PV + VCR Chillers	Solar-electric drive to power inverter-based chillers	Smart grid integration; zero carbon electricity
Solar Thermal + Absorption Chiller	Solar-thermal collectors provide heat to drive absorption	Off-grid viability; ideal for net-zero buildings
Grid-Responsive Chillers with TES	Load shifting using time-of-use tariffs and PCM/water storage	Reduces peak demand; enhances grid flexibility
Thermal Energy Storage (TES)	Stores cooling energy via chilled water or phase change	Enables load shifting; improves chiller efficiency
AI-Optimized Chillers	Uses IoT and predictive control for real-time optimization	Digital twin readiness; remote diagnostics



a) Solar photovoltaic-assisted chillers



b) Thermal Energy Storage Powered Vapour Absorption System

Fig. 2.9, Renewable-Integrated and Smart Chiller Systems

vi. Emerging Technologies

Several advanced cooling systems are under development globally, holding promise for future applications (Table 2.14).

These include:

- **Magnetocaloric cooling** works on the principle that certain materials change temperature when exposed to a magnetic field. This technology provides solid-state cooling without refrigerants and has the potential for high efficiency, though it is still limited to laboratory and pilot-scale demonstrations due to material and cost challenges.
- **Electrocaloric and thermoelectric cooling** both rely on solid-state phenomena, where electric fields or charge transport generate a temperature difference. These systems are compact, refrigerant-free, and environmental friendly, but their efficiency remains relatively low, and high material costs hinder wider adoption.
- **Liquid desiccant systems** use salt solutions to absorb moisture from the air while also reducing its temperature. They provide simultaneous dehumidification and cooling, making them especially effective in humid climates. However, they require periodic regeneration of the desiccant and are most practical in hybrid or solar-assisted systems.
- **Photonic and metamaterial-based radiators** are advanced engineered surfaces designed to optimize radiative heat rejection through the atmospheric window. They enable passive cooling by dissipating heat directly to the sky, reducing dependence on mechanical systems. While promising for building integration, their large-scale deployment is still in the emerging stage.

While still at the research or pilot stage, these technologies represent the next frontier in clean and compact cooling.

Table 2.14, Emerging Innovations in Chillers

Technology	Working Principle	TRL	**Notes**
Magnetocaloric Cooling	Uses magnetic field to change entropy of material	4–6	Solid-state, high potential efficiency
Electrocaloric Cooling	Electric field-driven entropy change	3–5	Under material and device optimization
Liquid Desiccant Cooling	Liquid salt absorbs moisture & cools air	6–7	Humid climates, commercial HVAC
Photonic Meta-Radiators	Engineered surfaces for superior radiation	4–6	Building skin integration

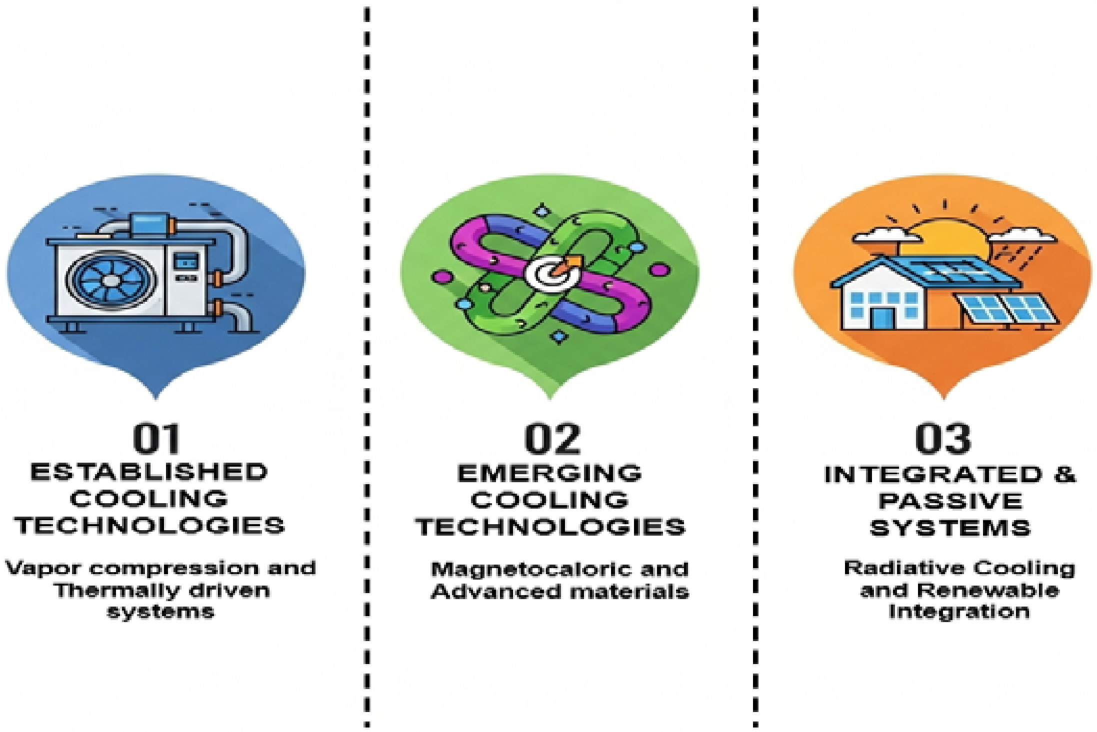


Fig. 2.10, Alternative chiller system innovations.

2.3 Current Infrastructure

The adoption of low-GWP refrigerants in the chiller sector depends not only on technical feasibility but also on a range of safety, regulatory, economic, and infrastructural factors. This section outlines the Technology Readiness Level (TRL) and key adoption barriers for each relevant refrigerant in the Indian context, including those recently introduced in international markets (Tables 2.15 and 2.16).

Table 2.15, Comparative Readiness and Barrier Matrix

Refrigerant/ Technology	Type	TRL	Current Use in India	Barriers	Remarks
R-1234yf / R-1234ze	HFO	TRL 8–9	Limited pilot-scale deployment	High cost, mild flammability (A2L), OEM dependency	Suitable for new chillers; global OEM support growing
R-290 (Propane)	Hydrocarbon	TRL 8	Used in small commercial systems	Highly flammable (A3), safety compliance	Excellent energy efficiency; building code restrictions apply
R-717 (Ammonia)	Natural	TRL 9	Common in industrial chillers	Toxicity, handling safety, zoning limitations	High COP; suited for large plant-scale cooling

R-744 (CO_2)	Natural	TRL 7–8	Pilot stage in cold chains	High pressure, complex system design	Potential for transcritical/ booster systems
R-32	HFC	TRL 9	Widely used in VRF and scroll chillers	Mild flammability (A2L); not ideal for very large chillers	Transitional refrigerant; moderate GWP
R-718 (Water)	Natural	TRL 6–7	Used in absorption chillers	Low cooling capacity, high footprint	Useful in solar/heat-driven cooling systems
R-444B	HFO-HFC blend	TRL 7–8	Emerging in global OEMs	Limited availability, A2L classification	Alternative to R-410A; not yet commercial in India
R-513A	HFO-HFC blend	TRL 8	Not yet deployed	Cost, availability	A1 classification; ideal retrofit for R-134a in water-cooled chillers
R-454B	HFO-HFC blend	TRL 8–9	Not commercial in India yet	Flammability (A2L), awareness gap	Lower-GWP substitute for R-410A; gaining global traction
R-452B	HFO-HFC blend	TRL 8	Very limited field reference	Flammability, limited OEM support	Drop-in replacement for R-410A; needs testing in Indian climates

i. Gaps and Barriers Identified

a Safety Standards and Regulatory Gaps

- Most refrigerants above fall under **A2L (mildly flammable)** or **A3 (highly flammable)** categories, which lack adequate recognition in **BIS safety codes**, **building bye-laws**, and the **National Building Code (NBC)**.
- **R-290** and **R-717** face specific deployment restrictions in occupied commercial spaces due to safety classification.

a. Technician Capacity and Training Deficit

- A large segment of India's HVAC-R workforce lacks formal training for:
  - ➢ Leak detection for flammable refrigerants
  - ➢ Emergency safety protocols
  - ➢ Pressure management in CO_2 systems
- Certification frameworks for A2L/A3 refrigerants are urgently needed.

a. Supply Chain and Component Availability

- **R-1234yf/ze**, **R-513A**, and **R-454B** are predominantly imported with limited domestic blending or manufacturing.
- Component compatibility (compressors, heat exchangers) for newer refrigerants is currently restricted to select OEMs.

a. Market Uncertainty and Financial Constraints

- Higher initial costs of low-GWP refrigerants and systems act as a deterrent in price-sensitive segments.

Lack of financial incentives, such as capital subsidies or preferential loans for low-GWP retrofits, limits uptake.

a. Retrofitting Limitations

- Direct retrofitting to R-513A (from R-134a) is feasible but needs capacity testing.
- R-454B and R-452B may require system redesigns or OEM-specific modifications for safe and efficient operation.

Table 2.16, Refrigerants Scalability Assessment

Parameter	High Scalability	Moderate Scalability	Low Scalability (Currently)
Policy Readiness	R-32, R-717	R-513A, R-1234ze	R-290, R-444B, R-744
Market Availability	R-32, R-717	R-1234yf, R-513A	R-444B, R-452B
Retrofit Compatibility	R-513A	R-32, R-1234ze	R-290, R-454B
Cost-Effectiveness	R-32, R-717	R-513A	R-1234yf, R-744

3. APPROACH AND METHODOLOGY FOR THE STUDY

This study adopted a structured methodology to evaluate the feasibility of introducing low-GWP refrigerant technologies in chillers for large air-conditioned buildings in India. The approach combined desk research, field inputs, technology benchmarking, policy and standards review, case study documentation, and stakeholder consultations. The steps followed are outlined below.

3.1 Methodology

A comprehensive review of current and emerging refrigerant and chiller technologies was undertaken to capture the available options and their potential application in India.

a) Technology Assessment

A comprehensive review of refrigerant and chiller technologies was undertaken.Non-ODS and low-GWP refrigerants such as HFOs (R-1234yf, R-1234ze, R-513A), natural refrigerants (R-717, R-290, CO_2, water), and transitional HFCs (R-32, R-452B, R-454B) were assessed.

- Performance parameters such as coefficient of performance (COP), part-load efficiency, and energy efficiency ratio (EER) were compared.
- Safety classifications (A1, A2L, A3, B2L) were mapped against system suitability for different types of chillers.
- Application mapping across centrifugal, screw, scroll, absorption, and hybrid chillers was carried out.

b) Safety and Standards Review

Safety and regulatory frameworks were examined at both national and international levels to determine readiness for next-generation refrigerants.

- Indian references included BIS codes (IS 16656, IS 16678, IS 16590, IS 18847) and BEE guidelines such as ECBC 2017.
- Global benchmarks such as ASHRAE-15, ISO 5149, and IEC 60335-2-40 were considered for comparison.
- The review highlighted gaps in Indian codes, particularly for mildly flammable and toxic refrigerants.

c) Challenges and Barriers

Barriers to adoption were identified through desk studies and direct consultations with stakeholders such as OEMs, facility managers, contractors, and policymakers.

- Supply chain gaps relating to refrigerant availability and component compatibility were identified.
- Market-level issues such as high initial costs and limited OEM deployment were analyzed.
- Skill gaps in handling flammable and toxic refrigerants were documented.

Regulatory bottlenecks arising from slow updates to codes and fragmented enforcement were highlighted.

- Limited awareness among end-users and decision-makers regarding long-term operational savings, total cost of ownership, and environmental compliance benefits.

d) Case Studies and Best Practices

A range of case studies, both international and Indian, were analysed to extract practical lessons for the transition.

- International cases included centrifugal chillers using R-1234ze in Europe, ammonia installations in industrial plants, and CO_2-based cascade systems in Japan.
- Indian examples included R-32 installations in Bengaluru, R-1234ze pilots in Pune, ammonia chillers in food processing units, and district cooling applications at GIFT City.
- Each case study highlighted key challenges, solutions adopted, and replicable practices.

d) Recommendations

Based on the findings, strategic recommendations were framed to accelerate the transition:

- Updating BIS and BEE codes to include provisions for low-GWP refrigerants.
- Introducing performance labelling for chillers.
- Developing training and certification programmes for technicians handling A2L and A3 refrigerants.
- Encouraging demonstration projects and public procurement of low-GWP systems.
- Linking refrigerant transition with green building certification schemes.
- Create financial instruments such as concessional loans or subsidies to address high upfront costs of low-GWP chillers.

f) Awareness Workshops

To ensure broad stakeholder involvement and validation of findings, two structured workshops were organized with the Ozone Cell, MoEF&CC:

- The first workshop conducted midway through the study, this session gathered views on technology readiness, safety considerations, and practical implementation challenges. Participants included OEMs, regulators, industry associations, and academic experts.
- The second workshop Organized towards the conclusion, this session presented the study's findings and recommendations to a wider group including policymakers, facility managers, manufacturers, and researchers. The workshop provided a platform to discuss actionable policy measures and align industry efforts with national priorities.

The methodology provided a structured, multi-dimensional evaluation of refrigerant technologies in India. By integrating a technology assessment, a safety and standards review, analysis of challenges and barriers, case studies, recommendations, and stakeholder workshops, the study developed a way forward for accelerating the transition to low-GWP chillers. This balanced approach ensures that future adoption in India will be technologically feasible, economically viable, and aligned with both environmental commitments and safety requirements[21].

Fig. 3.1, Approach and methodology on low-GWP alternative technologies for chillers in large-scale air-conditioning buildings.

4. EMERGING ALTERNATIVE LOW GWP TECHNOLOGIES

4.1 Low GWP options for chillers

India's adoption of non-ODS and low-GWP chiller technologies remains in a transitional phase. While conventional high-GWP refrigerants such as HCFC-22 and HFC-134a continue to dominate legacy systems, significant progress has been made in demonstrating and deploying alternative refrigerants in specific applications. A notable feature of this transition is the sectoral application of low-GWP options in chillers, commercial establishments such as offices, shopping malls, hospitals, and airports are increasingly adopting R-32, R-1234ze, and advanced HFO blends; industrial and cold chain sectors continue to depend on ammonia (R-717) and CO_2 systems; while institutional and public infrastructure, including universities and government campuses, are turning towards absorption and hybrid chillers powered by waste heat or solar energy. These sector-specific pathways underline that adoption is shaped by building type, operational load profiles, and safety requirements. The adoption landscape can therefore be understood in the following segments:

i. Commercial Sector Deployment

- **R-32-Based Scroll and Modular Chillers**:
  - ➢ Rapid uptake in high-end commercial office complexes, premium residential buildings, and institutional campuses.
  - ➢ Preferred due to its lower GWP (~675), availability, and energy efficiency improvements over R-410A.
  - ➢ Major OEMs offer R-32 chillers suitable for Indian conditions.
  - ➢ Increasingly integrated with smart Building Management Systems (BMS) to enhance load balancing and COP.
- **R-1234ze in Centrifugal Chillers**:
  - ➢ Pilot-scale adoption in metro cities (e.g., Mumbai, Delhi NCR, Bengaluru) primarily through international technology providers.
  - ➢ Deployed in large commercial campuses and green-certified buildings.
  - ➢ Supported by global sustainability certifications (LEED/GRIHA) and state-level energy conservation mandates.
- **Hydrocarbon-Based (R-290) Chillers**:
  - ➢ Limited deployment in small-capacity systems (<100 TR), particularly for standalone or rooftop installations.
  - ➢ Safety concerns due to A3 flammability rating have restricted widespread use in densely populated urban locations.

Deployed selectively in regions with strong safety enforcement frameworks and trained technical personnel.

- **R-454B, R-452B, and R-444B Blends**:

- These A2L-class HFO blends are being evaluated by selected OEMs as drop-in or new system refrigerants.
- R-454B and R-452B are positioned as replacements for R-410A, offering reduced GWP and comparable efficiency.
- Adoption is limited but growing in new commercial chillers where flammability mitigation infrastructure exists.

- **R-513A**:
  - A non-flammable A1 blend gaining interest as a replacement for R-134a in large capacity centrifugal chillers.
  - Offers lower GWP (~573) with no significant system redesign, making it viable for both retrofit and new systems.

ii. Industrial and Cold Chain Sector

- **Ammonia (R-717) Chiller Systems**:
  - Extensively used in cold storages, dairies, ice plants, fertilizer units, textiles, petrochemical complexes, and oil refineries.
  - High thermodynamic efficiency with zero GWP, making it the most sustainable refrigerant for large-scale industry.
  - Requires strict safety measures due to toxicity, including ventilation systems and trained operators.
- **CO_2-Based Cascade Systems (R-744)**:
  - Applied in pharmaceutical cold chains, dairy processing, frozen warehouses, and medical logistics.
  - Often used in cascade with ammonia or hydrocarbons to achieve ultra-low temperatures.
  - Advantages include very low GWP, but challenges are high pressure requirements and complex controls.
- **Hydrocarbons (R-290, R-600a, R-1270)**
  - Adopted in chemical plants, medium-capacity refrigeration, and some petrochemical facilities.
  - Propane (R-290) and isobutane (R-600a) are suited for smaller systems; propylene (R-1270) is relevant for petrochemical processes.
  - Provide excellent energy efficiency but face restrictions due to flammability (A3 classification).
- **HFOs and Blends (R-1234ze, R-514A, R-1233zd, R-1234yf)**
  - Emerging in pharmaceutical facilities, refineries, and petrochemical plants.
  - Ultra-low GWP (<10) combined with strong part-load efficiency.
  - Seen as long-term substitutes for R-134a in screw and centrifugal chillers.
- **Next-Generation HFCs and Blends (R-513A, R-452B, R-454B, R-407F)**
  - Transitional solutions where natural refrigerants face safety or infrastructure barriers.

- R-513A is used as a near drop-in for R-134a in pharma and industrial HVAC systems.
- R-452B and R-454B are suitable for process cooling in refineries and petrochemical facilities.
- Moderate GWP, but provide retrofit pathways for legacy systems.

- **Absorption Chillers (Ammonia–Water, LiBr–Water)**
  - Common in refineries, cement plants, petrochemical complexes, and industrial campuses.
  - Ammonia–water systems provide sub-zero cooling for process industries.
  - LiBr–water systems are favored in pharma R&D centers, hospitals, and campuses using waste heat or steam.
- **Cryogenic and Specialized Refrigerants (Ethylene, Ethane, Methane, R-23, R-508B)**
  - Applied in olefin plants, petrochemical crackers, LNG liquefaction, and refinery operations.
  - Ethylene (R-1150) and ethane (R-170) are standard in petrochemical refrigeration cycles.
  - R-23 and R-508B are used in pharmaceutical stability chambers and specialty chemical facilities requiring ultra-low temperatures.
  - Though some are legacy systems, they remain critical in niche industrial applications.

iii. Institutional and Public Infrastructure

- **Absorption Chillers (LiBr-Water)**:
  - Prominent in public hospitals, universities, and industrial campuses with access to waste heat, biomass, or solar thermal inputs.
  - Adopted under schemes supported by the Ministry of New and Renewable Energy (MNRE), especially in Smart Cities and SEZs.
  - Ideal for grid-deficit regions due to non-electricity driven operation.
- **District Cooling Initiatives**:
  - Emerging interest in low-GWP chillers as part of centralized cooling utilities in cities like Amaravati, GIFT City, and Kochi.
  - Preference for water-cooled centrifugal or screw chillers with R-1234ze, R-513A, or HFO-blends due to energy efficiency and low lifecycle cost.
  - Integrated with thermal energy storage (TES) and treated water reuse systems.

India's adoption of low-GWP chiller technologies is expanding through both top-down (policy, regulation) and bottom-up (market and technological readiness) approaches. The commercial sector shows promising momentum in deploying R-32 and HFO-based chillers, while ammonia and absorption technologies continue to dominate the industrial segment. The inclusion of newer blends such as R-454B, R-513A, and R-444B signals a widening portfolio of technically viable refrigerants. However, large-scale transformation will require synchronized efforts across regulatory reform, supply chain development, financial support mechanisms, and institutional capacity building.

4.2 Adoption of Technologies, Equipment Changes, and Retrofit Feasibility

Across the globe, the transition from ozone-depleting substances (ODSs) and high-GWP refrigerants to more climate friendly alternatives has become a defining feature of national cooling strategies. This shift is particularly visible in the deployment of large-capacity chillers that serve critical infrastructure such as airports, commercial complexes, hospitals, and data centers. Given the environmental urgency stemming from global warming and ozone layer depletion, countries are accelerating efforts to integrate low-GWP refrigerants and adopt alternative chiller technologies that align with their energy and climate commitments.

i. Technology Maturity and Deployment Readiness Matrix

A wide range of low-GWP and energy-efficient cooling technologies have emerged over the past decade to support the transition from conventional high-GWP refrigerant-based systems. These technologies vary in their level of technical maturity, commercial availability, and readiness for large-scale deployment, particularly in the Indian context. This section provides a structured assessment of these alternatives, focusing on their technological readiness and their applicability within India's built environment and policy framework. Advanced vapor compression systems using low-GWP refrigerants such as R-290 (propane), R-1234ze, and ammonia (R-717) have reached full commercial maturity and are already in active use in several commercial and institutional buildings across India. These systems, which include inverter-driven scroll and screw chillers, as well as high-efficiency centrifugal chillers with magnetic bearing compressors, are rated at Technology Readiness Level (TRL) 9. They offer high energy efficiency and compatibility with India's Energy Conservation Building Code (ECBC+) and green rating systems such as GRIHA. Their deployment is increasing, especially in new buildings and retrofit projects aiming for high energy performance (Table 4.1).

Thermally driven systems, including absorption and adsorption chillers, are also well-established. Absorption chillers based on lithium bromide-water and ammonia-water working pairs typically operate using waste heat or solar thermal energy. These systems, commonly used in hospitals, government campuses, and process industries, have reached TRLs of 7 to 9 and are commercially available in India. Adsorption chillers using silica gel or zeolite as solid desiccants are slightly less mature but are still deployable under pilot conditions, particularly in buildings with solar-assisted systems. Steam-ejector-based chillers, while less common, are beginning to find relevance in industries where low-grade steam is available, although they remain at TRLs of 6 to 7. Passive and hybrid cooling systems represent a promising direction for climate-adapted infrastructure. Technologies such as daytime and nighttime radiative cooling panels, cool roofs, and reflective coatings have reached TRLs between 6 and 8. While their cooling capacity may be modest compared to mechanical systems, they can play a significant role in reducing peak cooling demand. In India, institutions like IIT Guwahati have developed radiative cooling solutions suited to local climates, but widespread adoption is still in its early stages. Evaporative cooling systems, both direct and indirect, are widely used in dry and semi-arid regions and are already deployed at scale in factories, warehouses, and transit hubs. Earth-air heat exchangers, which utilize underground temperatures to cool incoming air, are also being tested in select campuses and data centers. Integrating renewable energy and storage with cooling systems is gaining traction. Solar photovoltaic or solar thermal systems paired with vapor compression or absorption chillers are being tested in public buildings and smart campuses. Thermal energy storage, using chilled water tanks or phase change materials, allows for load shifting and demand management, particularly during peak hours. These hybrid systems generally have TRLs between 7 and 8. Additionally, digital technologies, such as AI-based controls and IoT-enabled chiller optimization, are becoming part of smart building systems. While still at an early stage in India, these solutions have strong potential in commercial real estate and institutional buildings [22].

Several emerging technologies, including magnetic refrigeration, electrocaloric systems, and photonic radiators, remain at lower TRLs (between 3 and 6). Magnetic cooling systems, which use solid-state materials that change temperature under a magnetic field, have demonstrated cooling capacities and temperature drops in controlled environments but are not yet field-deployable in India. Photonic radiators and electrocaloric devices are under active research, with potential for integration into building skins or compact cooling devices, but their commercial readiness remains years away. District cooling systems, which aggregate cooling loads across multiple buildings, have already achieved TRL 9. A prominent example is the GIFT City project in Gujarat, which uses a central plant to provide chilled water to a mixed-use urban area. These systems offer economies of scale, higher energy efficiency, and the potential to incorporate low-GWP technologies at a district level. They are particularly suitable for SEZs, IT parks, transit hubs, and smart city developments.

In summary, technologies such as low-GWP vapor compression systems, absorption chillers, and evaporative cooling solutions are ready for immediate scale-up. Radiative and hybrid cooling systems are moderately mature and require demonstration projects and supportive policies to encourage adoption. Advanced solid-state technologies, while promising, remain in the research and development phase and would benefit from public investment and academic-industry partnerships. Efficient and sustainable cooling in the country can be accelerated by aligning technology deployment with national programs such as the India Cooling Action Plan (ICAP), Smart Cities Mission, and state-level green building mandates.

Table 4.1, Technology Maturity & Deployment Readiness Matrix

Category	Example Technologies	TRL	India Deployment Readiness
Advanced VCR – Low-GWP Refrigerants	Inverter scroll/ screw with R290, R1234ze, R717	9	Commercially viable; already in use for retrofit and new-build under ECBC+/GRIHA standards.
	Magnetic levitated centrifugal chillers with R1234ze	8	Ready abroad; early pilots possible in India (e.g., data centers, hospitals).

Thermally Driven Systems	LiBr-Water & NH_3-Water absorption chillers	8	Widely used in district cooling, hospitals, process industries—well supported by India's factsheet.
	Adsorption chillers (silica gel/zeolite)	7	Small-scale pilots in solar-thermal buildings; scaling projected by policy incentives.
	Steam-ejector chillers	6–7	Prototype/demo phase in industrial settings; best for site-specific waste-heat recovery.
Passive/ Hybrid Cooling	Radiative cooling panels (roof/ facade)	6–8	IIT Guwahati-developed coating prototype exists, but wide rooftop rollout pending. IIT Hyderabad came up with a concept of radiative cooling employing rooftop chilled water.
	Evaporative & IEC systems	7–9	Proven in dry/hybrid climates; operational in factories, large warehouses.
	Earth-air heat exchangers (EAHX)	6–7	Limited early trials in campuses/data centres; needs govt support for wider uptake.
Renewable + Storage Integration	Solar PV/T + VCR/ Thermal systems; TES with PCM	7–8	Pilot projects underway in smart campuses, metro stations, hospitals; NCAP aligned.
	Grid-responsive chillers (AI/IoT optimized)	7	Early integration trials in commercial buildings; high potential for smart city adoption.
Emerging Solid-State Technologies	Magnetocaloric cooling units	6	TRL 6 prototype tested in lab environment; Indian trials not yet. |
	Electrocaloric/ Photonic radiators	3–4	Early R&D stage; suitable for DST funded innovation projects.
District Cooling with VCR or Absorption	District cooling (e.g., GIFT City 10,000 TR)	9	Operational in GIFT City; scalable to other SEZs/smart cities.

ii. International Trends

Many countries have taken significant regulatory and policy steps to limit or ban the use of high-GWP refrigerants. For instance, the European Union's F-Gas Regulation aims to phase down HFC use by 79% by 2030, and has already imposed bans on refrigerants with a GWP greater than 2500 in certain systems (Table 4.2 and Fig. 4.1). The United States, through its AIM Act, is implementing a similar phasedown strategy, while several states have introduced GWP-specific restrictions and provided utility rebates for low-GWP alternatives.

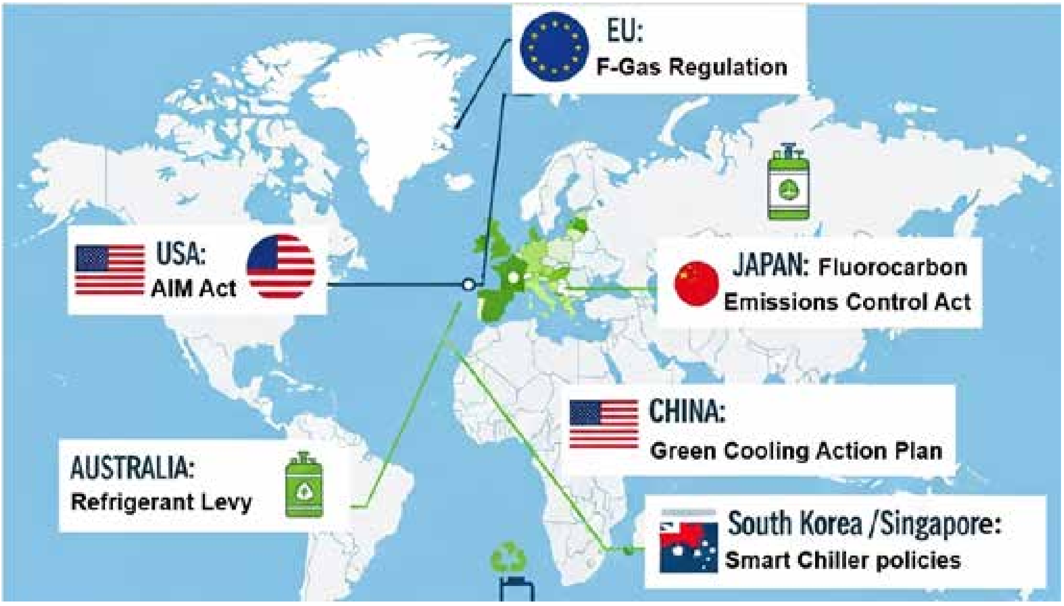


Figure 4.1: Global regulatory and frameworks for phasing out high-GWP refrigerants.

Japan has been integrating natural refrigerants such as CO_2 and ammonia in commercial chillers, supported by strong government-industry collaboration and a robust regulatory framework. China, on the other hand, has launched demonstration projects and pilot programs through its Green Cooling Initiative, aimed at enabling early-stage adoption of low-GWP refrigerants such as R-290 and R-717. Australia has adopted a refrigerant levy system and continues to promote the use of natural and HFO-based refrigerants in commercial cooling. These policy developments demonstrate a common trajectory: a gradual yet firm transition away from high-GWP substances, driven by a combination of legislative mandates, financial incentives, and awareness initiatives.

Table 4.2, Global Regulatory and Policy Frameworks Supporting Transition

Region / Country	Key Regulatory and Policy Instruments
European Union	The F-Gas Regulation mandates an 80% reduction in HFC use by 2030. Progressive bans on high-GWP refrigerants (>2500 GWP) have accelerated the uptake of natural refrigerants (NH_3, CO_2) and HFOs (R-1234ze).
United States (EPA & States)	Under the AIM Act, the U.S. targets an 85% phasedown of HFCs by 2036. ASHRAE has updated its codes (15 & 34) to accommodate flammable (A2L) refrigerants. States like California enforce GWP limits for chillers.
Japan	Known for early adoption and development of CO_2 and HFO-based chillers, with regulations under the Fluorocarbon Emissions Control Act. Offers public support for product innovation.
China	Through its Green Cooling Action Plan and partnership with international agencies (UNEP, UNIDO), China promotes pilot programs for low-GWP chillers, including R-290 and ammonia-based systems.
Australia	Implements a refrigerant levy and a cap-and-reduce mechanism, favoring R-1234yf/ze and natural refrigerants for large commercial applications.
South Korea & Singapore	Both countries support smart chiller technologies with low-GWP refrigerants and integrate cooling policies into building energy codes and green certification systems.

iii. Refrigerant Transitions and Global Technology Adoption

Several countries have prioritized the adoption of newer refrigerants that offer improved environmental profiles without compromising system performance. Hydrofluoroolefins (HFOs), particularly R-1234yf and R-1234ze, are being increasingly used in centrifugal and screw chillers across Europe, the United States, and parts of Asia. These refrigerants offer very low GWPs (typically less than 10), and while they are classified as mildly flammable (A2L), recent updates to building codes and safety standards have enabled their safe use in large-scale systems (Fig. 4.2). Ammonia (R-717) continues to be widely adopted in high-capacity industrial and district cooling applications, especially in colder regions of Europe and North America. It offers excellent energy efficiency and zero GWP but requires strict safety protocols due to its toxicity. Similarly, CO_2 (R-744) has gained traction in certain applications such as hospitals and data centres, though its high operating pressure and reduced performance in warmer climates have limited its use in tropical regions. Hydrocarbons like R-290 and R-600a are also being used in small-to-medium chiller systems, particularly in Europe. Despite their excellent thermodynamic properties and very low GWPs, their high flammability necessitates specialized handling, limiting their use in dense urban environments. HFC-32, a transitional refrigerant with a moderate GWP of around 675, has also found application in medium-capacity chillers, particularly in Asia. While it does not meet the long-term environmental targets under Kigali, it is considered an interim solution until natural refrigerants or ultra-low GWP blends become more commercially viable.

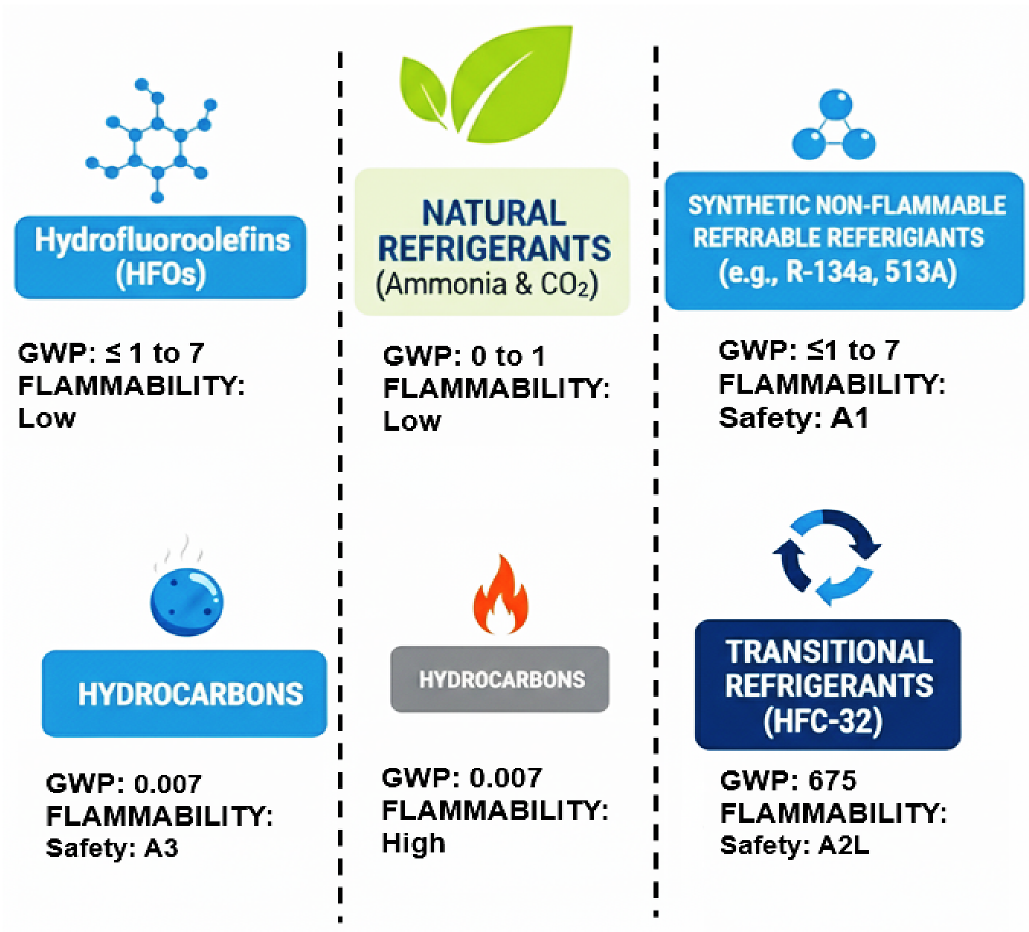


Figure 4.2: Adoption of different refrigerants globally.

iv. Alternative Chiller Systems: Beyond Conventional Refrigerants

In addition to replacing refrigerants, several regions are exploring alternative chiller system designs that eliminate or reduce reliance on vapor compression mechanisms altogether. Absorption chillers, which utilize thermal energy sources such as waste heat or solar thermal input, are widely adopted in Japan, Germany, and South Korea. These systems often use lithium bromide–water or ammonia–water pairs and are ideal for integration with combined heat and power (CHP) or district energy networks. While their coefficient of performance (COP) is lower than that of compression systems, their ability to function without electricity makes them a sustainable option in certain contexts. Hybrid systems, which combine absorption chillers with vapor compression or integrate thermal energy storage, are being piloted in Singapore and parts of the U.S. These systems are designed to reduce peak electricity demand and improve overall cooling efficiency. Radiative and evaporative cooling systems are also gaining attention in hot-arid regions such as the Middle East and southwestern United States. These systems offer passive or semi-passive pre-cooling, which, when integrated with conventional chillers, can significantly lower energy consumption. Some advanced economies are also investing in magnetic refrigeration and other solid-state cooling technologies. While these systems are still in the R&D stage, they offer the promise of eliminating refrigerants altogether, using magnetocaloric effects to achieve cooling.

v. Market Enablers and Implementation Mechanisms

The adoption of low-GWP refrigerants and alternative chiller systems around the world has been shaped not only by regulatory mandates but also by a range of supportive mechanisms introduced by governments, utilities, and industry bodies. These enablers play a crucial role in creating a favourable ecosystem for long-term market transformation and operational sustainability.

Several countries have aligned their refrigerant transition strategies with broader environmental and energy efficiency goals, integrating them into public procurement policies, financial incentive structures, and infrastructure modernization programs. In addition to technology readiness, emphasis is increasingly being placed on smart controls, modular deployment, and linkage with low-carbon district cooling systems. A few notable global practices are outlined below (Table 4.3)[23-26].

Table 4.3, Global Market Trends and Enablers

Trend	Examples
Digitalization and Smart Controls	Building Energy Management Systems (BEMS) equipped with AI-based optimization tools are widely used in the U.S. and European Union to monitor and improve chiller performance.
Modular Chiller Plants	Japan and South Korea have introduced scalable modular ammonia-based chiller units, which offer flexibility for phased expansion and simplified maintenance.
Green Public Procurement (GPP)	The European Union has made it mandatory for public buildings to prioritize low-GWP chillers under GPP directives, thereby stimulating demand through institutional projects.

Incentive Programs	Several U.S. states and EU member countries provide rebates, tax relief, or low-interest financing for businesses replacing high-GWP chillers with low-GWP models.
Integration with District Cooling	Countries like Qatar, Singapore, and Sweden are deploying HFO- and ammonia-based chillers as part of centralized district cooling networks, reducing peak electricity demand and emissions.

vi. Takeaways and Relevance for India

The global experience offers valuable insights for India's ongoing transition. Notably:

Interim targets have proven essential to guide industry response.

- Regulatory updates, especially around safety standards and refrigerant classifications, have enabled the use of mildly flammable refrigerants in buildings.
- Public sector pilot projects and demonstration facilities play a critical role in validating new technologies under local conditions.
- Financial incentives and training programs are necessary to address the dual challenges of cost and workforce readiness.
- Integration with building energy codes and green rating systems further accelerates adoption, especially in commercial buildings with high occupancy and cooling loads.

India can utilize these lessons by establishing a clear national strategy for refrigerant transition, supported by regulatory reform, demonstration programs, and targeted capacity-building initiatives.

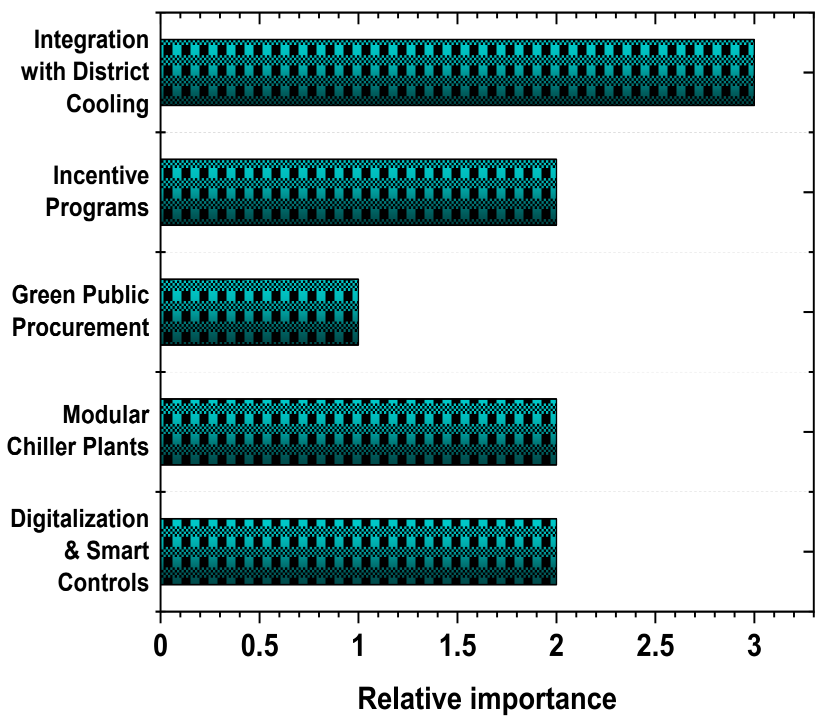


Figure 4.3, Lessons for India's refrigerant transition.

4.3 Supply Chain and Adoption Challenges

The transition to low-GWP refrigerant-based chillers in India is not only a technical shift but also a complex exercise in reconfiguring supply chains, industry readiness, workforce capabilities, and regulatory frameworks. Preliminary insights gathered from field consultations, secondary data, and stakeholder perspective indicate that India faces several distinct barriers that need to be addressed for wider adoption of climate-friendly cooling technologies.

i. Refrigerant Availability and Import Dependency

In India, the supply of low-GWP refrigerants such as HFO-1234yf, HFO-1234ze, R-290 (propane), and R-744 (CO_2) remains limited (Fig. 4.4). While domestic production of conventional HFCs is well established, there is a clear gap in the local manufacturing of newer alternatives. Most low-GWP refrigerants are still imported, which exposes users to price volatility, international regulatory constraints, and logistics delays. Additionally, due to limited demand, import volumes remain low, leading to supply chain fragmentation and lack of availability in regional markets beyond Delhi, Mumbai, Chennai, and Bengaluru.

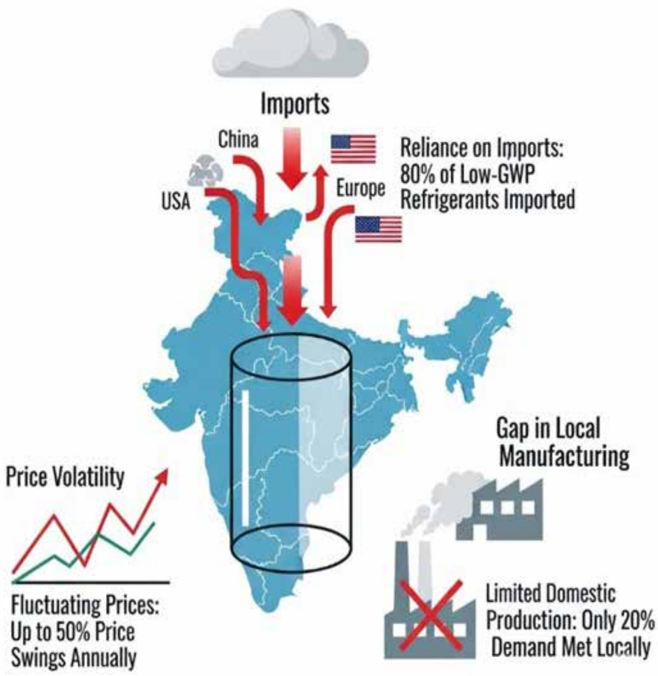


Figure 4.4 : Limited supply of low-GWP refrigerants in India.

ii. Limited Ecosystem for Compatible Equipment

Indian chiller manufacturers have taken initial steps to design systems compatible with low-GWP refrigerants, but the supply of critical components like compressors, expansion valves, and safety sensors is still dependent on global OEMs. Indigenous development of these components is at a nascent stage, and retrofitting legacy systems often becomes unfeasible due to a lack of backward compatibility (Fig. 4.5). For example, high-pressure CO_2 systems require entirely different design architectures, which are currently not supported by the mainstream supply chain in India.

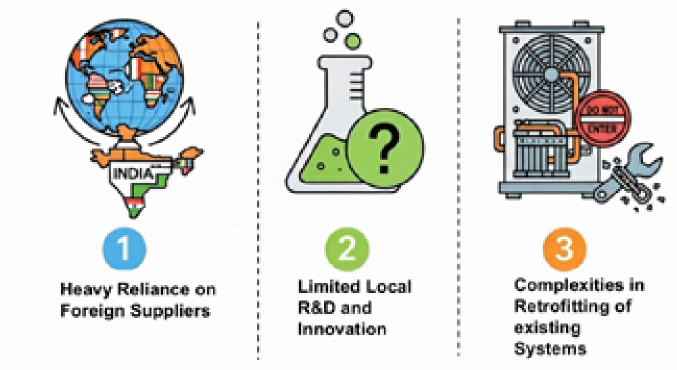


Figure 4.5: Indian chiller industry challenges.

iii. Skill and Knowledge Gaps Among Technicians

India's cold chain and HVAC-R workforce, while large, is largely trained to work with conventional refrigerants like R-22 and R-134a. Flammable (A3) and mildly flammable (A2L) refrigerants pose new challenges in terms of safety, handling, and leak detection. However, there are very few certified training centres that offer modules specifically for handling HFOs or natural refrigerants like ammonia and hydrocarbons . This creates reluctance among facility owners and maintenance contractors to adopt these systems. The absence of a national certification framework further limits confidence in technician capability.

iv. Economic Barriers and Market Readiness

Cost remains a major deterrent in the Indian context. Low-GWP chillers are currently 20–40% more expensive than their conventional counterparts. Without financial incentives or policy mandates, commercial building developers are hesitant to adopt them, especially in cost-sensitive sectors like hospitality, retail, or healthcare. While the long-term benefits through improved energy efficiency and carbon savings are clear, most users are guided by initial capital costs and shorter payback periods. In the absence of strong market demand, manufacturers also hesitate to scale production, keeping the technology in a low-volume, high-cost loop.

v. Gaps in Logistics and Cold Storage Infrastructure

Transporting flammable or toxic refrigerants safely requires specialized containers, trained logistics handlers, and adherence to safety regulations. India's logistics sector, especially in Tier-2 and Tier-3 cities, lacks the necessary infrastructure and awareness. During consultations, refrigerant suppliers pointed out that transport and insurance regulations for newer refrigerants are unclear, vary by state, and often cause shipment delays.

vi. Standards and Regulatory Alignment

The Bureau of Indian Standards (BIS) and the Bureau of Energy Efficiency (BEE) are already advancing work on updating relevant codes, and this creates a timely opportunity to strengthen the adoption of refrigerants such as R-290 and R-1234yf. Ensuring that fire safety clearances for A2L and A3 refrigerants are applied consistently across states will build confidence among users and regulators alike. Clear central guidance on integrating these refrigerants into public building codes and green rating systems such as GRIHA and ECBC will further accelerate uptake. In addition, a supportive public procurement framework that prioritizes low-GWP technologies in government projects could send a strong market signal and encourage faster, large-scale adoption.

5. CASE STUDIES: INTERNATIONAL AND NATIONAL

The case studies compiled to provide a detailed account of how low-GWP refrigerant technologies have been applied in large air-conditioning and district cooling systems across both international and Indian contexts (Tables 5.1 and 5.2). Each example outlines the refrigerant and technology employed, the best practices adopted, the challenges faced, and the solutions implemented. Together, these cases illustrate practical experiences, policy relevance, and technological pathways that can guide the wider adoption of low-GWP alternatives in different markets[27-29].

Table 5.1, International Case Studies on Low-GWP Refrigerant Adoption in Large Air-Conditioning Buildings

Project / Location	Refrigerant & Technology	Best Practices Highlighted	Key Challenge	Solution Approach
Gatwick Airport, UK	R-1234ze, inverter-controlled chillers	Low GWP, inverter control, BMS integration	Operational constraints in airport	Compact, efficient, and controlled installation
Pharmaceutical Facility, Austria	R-1234ze screw inverter chillers	High efficiency with reduced noise levels	Variable load and noise limits	Use of inverter control and low-noise design
Hospital Refurbishment, Spain	R-1234ze centrifugal chillers	Emission reductions and system reliability	Critical system cooling demand	Eco-friendly refrigerant with reliable compressor technology
Pharmaceutical Plant, Italy[28]	R-1233zd(E) with heat recovery	High COP, dual heating/cooling capability	Retrofit integration with heat systems	Smart controls with integrated heat recovery
Channel Tunnel, Europe	R-1233zd(E), high-capacity chillers	Non-flammable, low-GWP refrigerant with large capacity	Infrastructure upgrade	Retrofit with efficient, compliant systems
United Kingdom (Prototype Plant)	R-1234ze(E) prototype system	High part-load performance	Material and pressure drop issues	Component redesign and seal optimization
Japan (Industrial Facility)	Multi-stage centrifugal, R-1234ze(E)	Wide range: –20 °C brine to 60 °C hot water output	Balancing safety with high performance	Multi-stage configuration with material optimization

Europe (Magnetic-Bearing Facility)	R-1234ze centrifugal with magnetic bearing	High efficiency with advanced compressor technology	Adapting design to low-GWP refrigerant	Integration of magnetic-bearing optimized chillers
Denmark (District Heating)	R-1234ze (HFO) with heat recovery system	Cost-effective decarbonization and energy reuse	Transitioning urban heat networks	Drop-in refrigerant with performance and safety optimization
Geneva, Switzerland	HFO-based chillers	~5% efficiency gain compared with R-134a	Seasonal performance validation	Field trials to confirm efficiency across operating cycles

Table 5.2, India-Specific Case Studies on Low-GWP Refrigerant Adoption in Large Air-Conditioning[30-32]

Location / Facility	Technology Variant	Refrigerant Type Adopted	Best Practices & Impact	Challenges Addressed
Pune – Commercial Facility	Air-cooled central chillers	Solstice ze (R-1234ze; HFO, GWP < 1)	First Indian chillers using ultra-low-GWP HFO; aligns with environmental goals and sustainability initiatives.	Phased out high-GWP HFCs; introduced safe A2L refrigerant technology.
GIFT City, Gujarat	Centralized District Cooling System (DCS)	R-134a (non-CFC refrigerant)	India's first operational DCS; enhanced efficiency and reduced peak grid demand via centralized cooling.	Tackles inefficiencies of decentralized building-level AC in urban areas.
Bengaluru – Office Building	R-32 chiller system	R-32 (HFC, medium-GWP)	Selected for energy efficiency and reduced environmental footprint; widely adopted in modern commercial HVAC.	Improves chiller environmental performance compared with older HFC systems.

Dahej, Gujarat – Refrigerant Facility	HFO refrigerant production plant	Solstice zd (R-1233zd; HFO, GWP ≈ 1)	Local production of ultra-low GWP refrigerant strengthens domestic supply chain and sustainability.	Reduces dependence on imported refrigerants; bolsters local availability.
Multiple Installations – Across India	Low-GWP chiller systems	Solstice zd (R-1233zd(E); HFO, GWP ≈ 1)	Early adoption of low-GWP refrigerants enhances India's chiller industry sustainability credentials.	Supports transition away from high-GWP HFCs; aligns with Kigali Amendment commitments.
National & Global Chiller Sector (India)	HFO-ready chiller systems emerging	R-513A & R-514A (HFO blends)	HFO blends identified as viable low-GWP alternatives for future district cooling and high-efficiency systems.	Prepares industry for transition from R-134a and supports future chiller design upgrades.

6. CURRENT INDIAN AND GLOBAL STANDARDS

India's transition to sustainable cooling technologies under the Montreal Protocol and the Kigali Amendment offers an excellent chance to modernize safety and performance standards. As low-GWP refrigerants such as HFOs, natural options, and advanced blends are introduced in place of high-GWP gases, a strong regulatory framework will ensure their safe and efficient use in chiller systems. The Bureau of Indian Standards (BIS) and the Bureau of Energy Efficiency (BEE) already play a key role in shaping these guidelines for large-scale air-conditioning applications.

BIS has issued important standards that provide direction on refrigerant classification, system design, and safe operation. IS 16572:2016, aligned with ISO 817, categorizes refrigerants based on flammability and toxicity, which is especially relevant for R-1234yf, R-290, and CO_2. IS 14785:2022 defines requirements for the design and installation of systems using A2L and B2L refrigerants. IS/ISO 817:2014 further supports system engineering and maintenance planning. At the same time, some older standards, such as IS 11329:1985 and IS 3034:1993, would benefit from updates so that they reflect advances in automation, digital controls, and the handling of newer refrigerants.

In parallel, BEE has advanced the agenda of energy efficiency through tools such as the Energy Conservation Building Code (ECBC). The 2017 version sets minimum performance values for chillers, including COP and Integrated Part Load Value (IPLV) benchmarks, while mechanisms like the Perform, Achieve and Trade (PAT) scheme provide additional incentives for energy savings. The Eco-Niwas Samhita adds provisions for mixed-use and residential projects where centralized cooling is becoming more common. The draft labelling program for chillers, once finalized, will create clear signals in the market for low-GWP and high-efficiency systems.

Moving forward, India has several opportunities to strengthen this progress:

- Revise BIS standards to fully include emerging refrigerants such as HFO blends and CO_2.
- Operationalize the chiller star-labelling program under BEE.
- Launch large-scale technician training and certification under Skill India, focusing on safe handling of A2L, A3, and B2L refrigerants.
- Integrate refrigerant safety and indoor air quality into green building frameworks like GRIHA and LEED.
- Develop a retrofit safety framework to guide the smooth phase-out of legacy refrigerants such as R-22 and R-134a.

By combining regulatory updates, training initiatives, and market incentives, India is well placed to lead the safe and sustainable adoption of low-GWP chiller technologies while meeting its climate commitments.

6.1 BIS Standards for Safety and System Design

The Bureau of Indian Standards (BIS) provides essential safety and performance guidelines for refrigerants and chiller systems used in commercial buildings. These standards are crucial for ensuring that low-GWP refrigerants, many of which are mildly flammable or operate at high pressures can be adopted safely in India's air-conditioning sector[33,34].

Some key standards relevant to low-GWP chillers include (Table 6.1):

- **IS 16656:2017 / ISO 817:2014 –** This standard assigns designation to refrigerants and classifies them by toxicity and flammability (e.g., A1, A2L, A3, B2L). It is the basis for handling newer refrigerants such as R-32, R-1234yf, R-290, or ammonia.
- **IS 16678 (Part 1):2018 / ISO 5149-1:2014 –** Specifies safety and environmental requirements for refrigerating systems and heat pumps. It is particularly important for installations using flammable (A2L/A3) or toxic (B2L) refrigerants.
- **IS 5610:2025 –** Refrigerants: Specification. Defines quality, purity, testing methods, and packaging requirements for refrigerants supplied in India. Ensures consistency in refrigerant supply for chillers and prevents contamination or unsafe mixtures.
- **IS 16590:2023 –** Liquid Chilling Package Units — Specification. Provides the product-level requirements and performance criteria for water-cooled and air-cooled chillers, applicable to low-GWP refrigerants.
- **IS 11327:2022 –** Requirements for refrigerant condensing units. Ensures safe and efficient operation of condensing modules, including those using low-GWP refrigerants.
- **IS/IEC 60335-2-40:2018 –** Safety requirements for heat pumps, air-conditioners, and dehumidifiers using flammable refrigerants. Useful where chillers interface with distributed HVAC equipment in large buildings.
- **IS 18847:2024 / ISO 22712:2023 –** Defines competence requirements for technicians handling refrigerants and servicing systems. Critical for safe adoption of A2L/A3/B2L refrigerants in India.

Table 6.1, BIS Standards Relevant to Chiller Systems

Standard	Title / Description	Applicability to Low-GWP Chillers
IS 16656:2017 / ISO 817:2014	Refrigerants — Designation and safety classification	Assigns refrigerants to safety classes (A1, A2L, A3, B2L); basis for handling mildly flammable and toxic refrigerants
IS 16678 (Part 1):2018 / ISO 5149-1:2014	Refrigerating systems and heat pumps — Safety and environmental requirements	Governs safe design, installation, and maintenance of systems using A2L, A3, and B2L refrigerants
IS 5610:2025	Refrigerants — Specification (Third Revision)	Sets purity, testing, and packaging rules; mandatory for refrigerant procurement and supply
IS 16590:2023	Liquid Chilling Package Units — Specification	Defines performance and safety requirements for water-cooled and air-cooled chillers
IS 11327:2022	Requirements for refrigerant condensing units	Ensures reliability and safety of condensing units across HFC, HFO, and natural refrigerants
IS/IEC 60335-2-40:2018	Household and similar electrical appliances — Safety for heat pumps, ACs, and dehumidifiers	Applicable to chillers interfacing with distributed equipment using flammable refrigerants
IS 18847:2024 / ISO 22712:2023	Competence of personnel for refrigerating systems	Ensures technicians handling low-GWP refrigerants are certified for safety and reliability

6.2 BEE Guidelines on Energy Performance

The Bureau of Energy Efficiency (BEE) has been instrumental in shaping India's energy efficiency framework for the building and HVAC sectors (Table 6.2). The Star Labelling Programme has achieved notable success in influencing the market for room air conditioners, but equivalent regulatory measures for large centralized chillers are still at an early stage. The Energy Conservation Building Code (ECBC 2017) provides benchmarks for chillers through requirements such as Coefficient of Performance (COP), Integrated Part Load Value (IPLV), and control strategies for optimized part-load operation. However, implementation differs across states and often depends on the enforcement strength of local authorities, resulting in uneven outcomes. In recent years, BEE has moved to address this gap with the Voluntary Labelling Programme for chillers, currently applicable to centrifugal and screw systems. Though not mandatory, this programme signals an important policy direction, encouraging building owners and developers to consider lifecycle performance in procurement decisions. Its design reflects India's climatic conditions, ensuring that efficiency benchmarks are aligned with real operating environments rather than laboratory-only performance.

Alongside, large projects such as airports, hospitals, IT parks, and shopping malls that run centralized chiller plants may fall under the Perform, Achieve and Trade (PAT) scheme if designated as energy-intensive consumers. PAT provides the dual benefit of compliance and financial incentive by allowing verified savings to be converted into Energy Saving Certificates (ESCerts) for trading. This market-linked mechanism supports investment in low-GWP, high-performance chillers. The Eco-Niwas Samhita (ENS), while primarily aimed at residential buildings, includes provisions for HVAC systems in shared spaces of large housing complexes. Although its coverage of centralized cooling is currently limited, future revisions are expected to broaden its scope in response to the rising demand for multi-unit residential cooling.

Taken together, ECBC, PAT, ENS, and the voluntary chiller labelling scheme provide a multi-pronged regulatory pathway for improving efficiency in the chiller sector (Fig. 6.1). Strengthening enforcement, widening the coverage of labelling, and linking these programmes

with financial incentives will be critical for accelerating the transition toward low-GWP and high-efficiency chiller technologies in India.

Figure 6.1: BEE guidelines for energy performance of chillers.

In summary, while key policies such as ECBC and PAT recognize the role of chillers in building energy performance, there is a need to:

- Operationalize the chiller labelling scheme,
- Standardize enforcement of ECBC provisions, and
- Encourage state-level adoption through integration with Urban Local Bodies (ULBs) and State Designated Agencies (SDAs).

These measures, when effectively implemented, can support the broader transition to energy-efficient, low-GWP refrigerant technologies in India's commercial cooling sector[35-38].

Table 6.2, BEE Policies Applicable to Chillers

Policy / Code	Scope	Relevance to Low-GWP Chillers
ECBC 2017 (Energy Conservation Building Code)	Sets energy performance limits (e.g., COP, IEER) for HVAC systems in commercial buildings	Applicable to new buildings; includes COP benchmarks for chillers
Voluntary Energy Labelling (Draft)	Performance labels for centrifugal/screw chillers	Notified but not implemented; would help differentiate efficient low-GWP systems

Eco-Niwas Samhita 2021 – Part II	Energy efficiency for residential/mixed-use buildings	Relevant where central chillers serve luxury residential projects
PAT Scheme	Market mechanism for energy-intensive sectors	Chillers can be considered for energy savings credits under this scheme

6.3 Identified Gaps in Standards and Guidelines

BIS and BEE have laid a strong foundation for safety and energy efficiency in HVAC systems, but further strengthening is needed to fully support the transition to low-GWP refrigerants. Insights from desk research and stakeholder consultations highlight a few critical areas where progress can be accelerated:

- **Coverage Gaps in BIS Standards:** While recent updates such as IS 5610:2025, IS 16656, IS 16678, and IS 16590 have brought refrigerant safety and chiller specifications closer to global best practices, several older codes remain misaligned with the properties of A2L, A3, and B2L refrigerants. Importantly, legacy standards still do not adequately cover HFO blends or CO_2-based systems, leaving room for revisions that can better reflect current technology trends.

- **Chiller Labelling Not Yet Mandatory:** In contrast to room air-conditioners, there is currently no compulsory star-labelling scheme for chillers. This absence makes it difficult for end-users and facility managers to compare performance levels or refrigerant options across equipment. Finalizing and notifying a performance label for chillers would create greater transparency and encourage adoption of efficient, low-GWP technologies.

- **Technician Training Still Limited:** IS 18847:2024 defines the competence requirements for refrigerant handling, but large-scale training and certification programs are yet to be implemented. As a result, many technicians continue to work with A2L, A3, and B2L refrigerants without adequate preparation, creating both safety and performance risks. A structured, nationwide training framework would help bridge this gap and ensure safe adoption of next-generation refrigerants.

- **ECBC Implementation:** ECBC 2017 sets benchmarks for efficiency, including performance requirements for chillers. However, enforcement remains uneven across states and urban local bodies. A uniform approach to implementation would improve consistency and amplify the benefits of the code.

Retrofit Guidance: Many R-22 and high-GWP HFC chillers are now being replaced, yet there are no standardized retrofit protocols to guide safe and effective conversions. Developing retrofit guidelines will ensure that transitions to low-GWP refrigerants are both efficient and safe, minimizing risks during replacement.

The gaps show the need for regular revision of BIS codes, early notification of BEE chiller labels, structured training for service personnel, and clear retrofit rules for transition to low-GWP systems (Fig. 6.2).

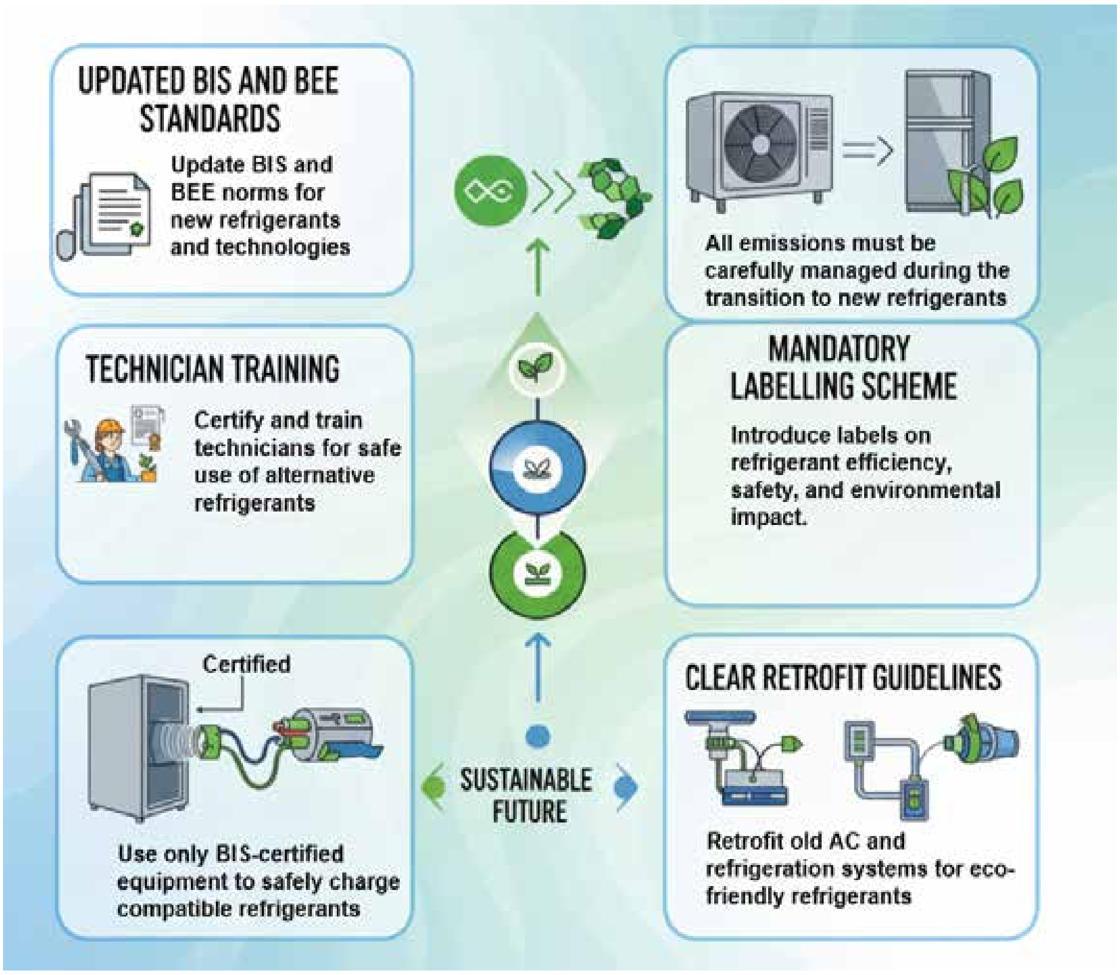


Figure 6.2: India's refrigerant transition.

7. KEY FINDINGS AND CONCLUSION

The adoption of low-GWP technologies in chillers is a global trend driven by HFC phase down schedule of the Kigali Amendment under the Montreal Protocol, aiming to reduce the climate Impact. Transitioning to low GWP options necessitates specialized equipment and potential higher costs, whereas new chillers are increasingly designed for HFOs, hydrocarbons (like propane), CO_2, and refrigerant blends, making low-GWP options a necessity for sustainable thermal comfort in large buildings.

The key findings from the study are as follows:

- Natural refrigerants offer strong climate benefits and competitive efficiencies: ammonia delivers high COP with zero GWP; propane combines very low GWP with good performance; CO_2 provides near-zero GWP and is proving valuable in specialised applications. HFOs combine high efficiency with ultra-low GWP and show promising part-load performance in centrifugal chillers. Pilot installations (e.g., R-1234ze) have demonstrated measurable part-load efficiency gains.
- In terms of technology readiness and availability, technically proven low-GWP refrigerants for chillers that are commercially available include natural refrigerants (ammonia R-717, hydro-carbon based propane R-290, CO_2 R-744), next-generation HFOs (e.g., R-1234ze, R-1234yf, R-513A), and transitional HFCs (e.g., R-32, R-452B).
- Adoption of low-GWP alternatives in India is gradually taking shape, with metro cities such as Delhi NCR, Mumbai, and Bengaluru leading the way through pilot projects using R-32, HFOs (R-1234ze, R-513A), and natural refrigerants in larger chillers. While penetration in tier-II and tier-III cities is currently below 10%, this presents a strong opportunity for scaling through supportive policy measures, promoting awareness and developing competencies of the personnel associated with maintaining and servicing, particularly handing the A2L and A3 refrigerants through unified certification and training programmes.
- In terms of retrofitting versus replacement of existing chillers, retrofitting provides a cost-effective solution, typically reducing upfront expenses by 20–30% for mid-life systems while improving efficiency and enabling a transition to low-GWP refrigerants. Replacements enhance safety and performance, ensure alignment with emerging standards and sustainable cooling objectives, and unlock significant long-term energy and emissions savings. A coordinated approach that combines targeted retrofitting for newer systems with proactive replacement of aging chillers can facilitate a smoother, more effective transition to a resilient, sustainable, and energy-efficient cooling sector in India.
- India's shift to low-GWP refrigerants faces some economic and supply challenges. Alternative refrigerants, like R-1234ze and R-454B, are imported, making them 15–25% more expensive and potentially harder to access. Expanding domestic manufacturing and refrigerant supply facilities can strengthen the supply chain, reduce costs, and ensure reliable availability, supporting wider adoption across the country.

Pilot projects in India have shown that HFO centrifugal chillers, such as R-1234ze, can achieve up to 15% higher part-load efficiency compared to conventional HFC-134a units, demonstrating clear environmental and operational benefits. Market surveys indicate that over 70% of facility managers are willing to adopt low-GWP refrigerants if supported by clear safety codes, reliable service, and financing incentives. Showcasing these benefits and market readiness can bridge the gap between technology availability and widespread deployment, enabling a faster and smoother transition in India's chiller sector.

- Updating safety standards for A2L and A3 refrigerants, strengthening domestic production and distribution of low GWP refrigerants, implementing demonstration projects, and public procurement can collectively drive wider market uptake and wider adoption of low GWP options.
- Raising awareness through capacity building programs and seminars among stakeholders including chiller manufacturers, HVAC associations, green building councils, facility managers, building developers, engineers, financial institutions, academia, and government officials will promote adoption of low GWP options in chillers.
- Integration of low-GWP chillers with renewable energy sources such as solar PV and thermal energy storage offers additional opportunities to cut peak electricity demand and align with India's clean energy goals.
- Digital tools such as IoT-based monitoring, predictive maintenance, and AI-driven controls have been shown to improve operational efficiency and provide early detection of refrigerant leaks, adding both safety and performance advantages.
- Sectoral adoption is diversifying, with commercial complexes adopting R-32 and HFO blends, industrial facilities and refineries continuing with ammonia, CO_2, and cryogenic refrigerants, and institutional campuses favouring absorption and hybrid systems powered by waste heat or solar energy.
- International experience underscores the importance of aligning policies, incentives, and procurement frameworks. Public sector projects in India have already demonstrated the viability of low-GWP chillers in Smart Cities, SEZs, and institutional campuses, showing strong potential for scale-up through public–private partnerships.
- Widespread adoption of low-GWP chillers can avoid millions of tons of CO_2 emissions across their lifetime while also improving energy security and air quality. Their deployment in public buildings, hospitals, and schools enhances social resilience, especially during extreme heat events.
- Establishing a long-term roadmap that links standards, financial incentives, training, and demonstration projects with the India Cooling Action Plan and national net-zero targets will ensure sustained progress and help position India as a leader in climate-friendly cooling technologies.

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